

This is a part of SFQED-Loops script collection
developed for calculating loop and tree-level processes in
Strong-Field Quantum Electrodynamics.

The scripts are available on <https://github.com/ArsenyMironov/SFQED-Loops>

If you use this script in your research, please, consider
citing our papers:

- A. A. Mironov, S. Meuren, and A. M. Fedotov, PRD 102, 053005 (2020); arXiv:2003.06909
<https://doi.org/10.1103/PhysRevD.102.053005>

- A. A. Mironov, A. M. Fedotov, 105 (3), 033005 (2022); arXiv:2109.00634
<https://doi.org/10.1103/PhysRevD.105.033005>

- Y. V. Selivanov, A. A. Mironov, A. I. Alexeenko, A. M. Fedotov arXiv:2605.12479 (2026)

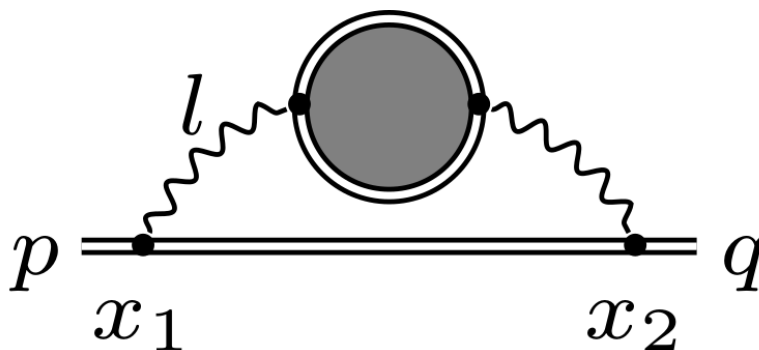
If you have any questions, please, don't hesitate to contact:
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```
In[1]:= NotebookEvaluate[NotebookOpen[NotebookDirectory[] <> "definitions.nb"]]
```

FeynCalc 10.1.0 (stable version). For help, use the online documentation, visit the forum and have a look at the supplied examples. The PDF-version of the manual can be downloaded [here](#).

If you use FeynCalc in your research, please evaluate FeynCalcHowToCite[] to learn how to cite this software.

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Mass operator ($e > 0$)

$$-iM(q, p) =$$

$$(ie)^2 \Lambda^{2D-8} \int d^D x_1 d^D x_2 \bar{E}_q(x_2) \gamma^\mu S^c(x_2, x_1) \gamma^\nu E_p(x_1) D_{\mu\nu}^c(x_1 - x_2);$$

x_1^μ, x_2^μ – position of the left and right vertices of the diagram;

p^μ, q^μ – initial and final electron momenta;

$S^c, D_{\mu\nu}^c$ – electron and photon casual propagators;

The Ritus E_p – function

$$E_p(x_1) = \left[1 - \frac{e(\gamma k)(\gamma a)}{2(kp)} (kx_2) \right]$$

$$\text{Exp} \left[-i(p x_1) + i \frac{e(a p)}{2(k p)} (k x_1)^2 + i \frac{e^2 a^2}{6(k p)} (k x_1)^3 \right];$$

$$\bar{E}_q(x_2) = \gamma^0 E_q(x_2) \gamma^0;$$

Electron propagator in a CCF in D dimensions

$$S^c(x_2, x_1) = e^{i\eta} S_{\text{diag}}^c(x_2 - x_1) =$$

$$= e^{i\eta} e^{-i\frac{\pi}{2}\left(\frac{D}{2}-1\right)} \frac{\Lambda^{4-D}}{2^D \pi^{D/2}} m$$

$$\int_0^\infty \frac{ds}{s^{D/2}} \left[1 + \frac{(\gamma x)}{2ms} - \frac{e^2 s (\gamma F F x)}{3m} + \frac{i}{2} e s (\sigma^{\alpha\beta} F_{\alpha\beta}) + \frac{i e F_{\alpha\beta}^* x^\beta \gamma^\alpha \gamma^5}{2m} \right] e^{-is - i\frac{x^2}{4s} + i\frac{s}{12} e^2 (Fx)^2}$$

$$= e^{i\eta} e^{-i\frac{\pi}{2}\left(\frac{D}{2}-1\right)} \frac{\Lambda^{4-D}}{2^D \pi^{D/2}} m$$

$$\int_0^\infty \frac{ds}{s^{D/2}} \left[1 + \frac{(\gamma x)}{2ms} + \frac{e(\gamma a)(kx)}{2m} - \frac{e(\gamma k)(ax)}{2m} + \frac{e(\gamma x)(\gamma a)(\gamma k)}{2m} + \frac{e s (\gamma a)(\gamma k) + \frac{e^2 a^2 s (\gamma k)(kx)}{3m}}{2m} \right] e^{-is - i\frac{x^2}{4s} + i\frac{s}{12} e^2 (Fx)^2}$$

$$\eta = e(ax)(k, (x_1 + x_2)/2),$$

$$x = x_2 - x_1,$$

$$e > 0,$$

$[\Lambda] = m$ – mass scale,

$$\sigma^{\alpha\beta} = \frac{i}{2} (\gamma^\alpha \gamma^\beta - \gamma^\beta \gamma^\alpha),$$

$$\gamma^5 = i\gamma^0 \gamma^1 \gamma^2 \gamma^3,$$

$$F_{\alpha\beta} = k_\alpha a_\beta - k_\beta a_\alpha,$$

$$F_{\alpha\beta} F^\beta{}_\lambda = -a^2 k_\alpha k_\lambda$$

$$e^{-i\frac{\pi}{2}\left(\frac{D}{2}-1\right)} \frac{\Lambda^{4-D}}{2^D \pi^{D/2}} m^{D-1} \rightarrow \frac{(-i) m^3}{16 \pi^2}, \quad D \rightarrow 4$$

Exact photon propagator in momentum representation

$$D_{\mu\nu}^c(l) =$$

$$D_0(l^2) g_{\mu\nu} + D_1(l^2, \chi_l) \epsilon_\mu^{(1)}(l) \epsilon_\nu^{(1)}(l) + D_2(l^2, \chi_l) \epsilon_\mu^{(2)}(l) \epsilon_\nu^{(2)}(l);$$

l^μ – the photon 4 – momentum;

$$\chi_l = \frac{e}{m^3} \sqrt{-(F_{\mu\nu} l^\nu)^2};$$

$$\epsilon_\mu^{(1)}(l) = \frac{e F_{\mu\nu} l^\nu}{m^3 \chi_l};$$

$$\epsilon_\mu^{(2)}(l) = \frac{e F_{\mu\nu}^* l^\nu}{m^3 \chi_l}; \quad F^{*\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\delta\lambda} F_{\delta\lambda};$$

$$(\epsilon^{(i)}(l))^2 = -1;$$

$$D_0(l^2) = \frac{-i}{l^2 + i0}, \quad D_{1,2}(l^2, \chi_l) = \frac{i\Pi_{1,2}}{(l^2 + i0)(l^2 - \Pi_{1,2})};$$

$\Pi_{1,2} = \Pi_{1,2}(l^2, \chi_l)$ – polarization operator eigenfunctions;

Exact photon propagator in coordinate representation

$$D_{\mu\nu}^c(x) = \frac{\Lambda^{4-D}}{(2\pi)^D} \int d^D l D_{\mu\nu}^c(l) e^{-ilx} =$$

$$= \text{Exp} \left[-i \frac{\pi}{2} \left(\frac{D}{2} - 2 \right) \right] \frac{1}{2^D \pi^{D/2}} \frac{\Lambda^{4-D}}{m^{2-D}}$$

$$\int_0^\infty \frac{dt}{t^{D/2}} e^{-i \frac{m^2 x^2}{4t}} \left\{ g_{\mu\nu} J_0(t) + \left(-2i t \frac{k_\mu k_\nu}{m^2 \phi^2} + \frac{e^2 F x_\mu F x_\nu}{m^2 \xi^2 \phi^2} \right) J_1(t, \chi_l) \right. \\ \left. + \left(-2i t \frac{k_\mu k_\nu}{m^2 \phi^2} + \frac{e^2 F D x_\mu F D x_\nu}{m^2 \xi^2 \phi^2} \right) J_2(t, \chi_l) \right\};$$

$$J_k(t, \chi_l) = -i \int_0^\infty dl^2 D_k(l^2, \chi_l) e^{-il^2 t};$$

$$\chi_l = \xi k l / m^2 = \xi \phi / 2 m^2 t;$$

Preliminaries

Let us define momenta and coordinate variables

$$X = x_2 - x_1;$$

$$X = \frac{1}{2} (x_1 + x_2);$$

$$\phi = kx;$$

$$\Phi = kX;$$

$$\phi_1 = kx_1;$$

$$\phi_2 = kx_2;$$

The functions `NewMomentum` and `NewCoordinate` are predefined in the file `definitions.nb`

They provide the corresponding 4 – vector along with all possible contractions with the field tensor $F_{\mu\nu}$ and 4 – vectors a_μ, k_μ (e.g. $Fx^\nu{}_{[\mu]} = (Fx)_\mu = F_{\mu\nu} x^\nu$, see more details in the file `definitions.nb`)

```
In[2]:= NewMomentum["p"]
NewMomentum["q"]
NewMomentum["l"]
NewCoordinate["x1"]
NewCoordinate["x2"]
NewCoordinate["x"]
NewCoordinate["X"]
ScalarProduct[k, x] =  $\phi$ ;
ScalarProduct[k, X] =  $\Phi$ ;
ScalarProduct[k, x1] =  $\phi_1$ ;
ScalarProduct[k, x2] =  $\phi_2$ ;
ScalarProduct[Fx, Fx] =  $-\xi^2 \phi^2 * m^2 / e^2$ ;
ScalarProduct[FDx, FDx] =  $-\xi^2 \phi^2 * m^2 / e^2$ ;
ScalarProduct[x, FFx] =  $\xi^2 \phi^2 * m^2 / e^2$ ;
```

$$\begin{aligned}
& \left\{ p^\alpha, p^2, k \cdot p, Fp^\alpha, FFp^\alpha, FDP^\alpha, a \cdot p, 0, 0, 0, -a^2 (k \cdot p), 0, 0, -\frac{m^6 \chi p^2}{e^2}, -\frac{m^6 \chi p^2}{e^2}, \frac{m^6 \chi p^2}{e^2}, 0, 0, 0, 0, 0, 0 \right\} \\
& \left\{ q^\alpha, q^2, k \cdot q, Fq^\alpha, FFq^\alpha, FDq^\alpha, a \cdot q, 0, 0, 0, -a^2 (k \cdot q), 0, 0, -\frac{m^6 \chi q^2}{e^2}, -\frac{m^6 \chi q^2}{e^2}, \frac{m^6 \chi q^2}{e^2}, 0, 0, 0, 0, 0, 0 \right\} \\
& \left\{ l^\alpha, l^2, k \cdot l, Fl^\alpha, FFl^\alpha, FDI^\alpha, a \cdot l, 0, 0, 0, -a^2 (k \cdot l), 0, 0, -\frac{m^6 \chi l^2}{e^2}, -\frac{m^6 \chi l^2}{e^2}, \frac{m^6 \chi l^2}{e^2}, 0, 0, 0, 0, 0, 0 \right\} \\
& \left\{ x1^\alpha, x1^2, k \cdot x1, a \cdot x1, Fx1^\alpha, FFX1^\alpha, FDx1^\alpha, k \cdot x1, 0, 0, 0, \right. \\
& \quad \left. -a^2 (k \cdot x1), 0, 0, -\frac{m^2 \xi^2 (k \cdot x1)^2}{e^2}, -\frac{m^2 \xi^2 (k \cdot x1)^2}{e^2}, \frac{m^2 \xi^2 (k \cdot x1)^2}{e^2}, 0, 0, 0, 0, 0, 0 \right\} \\
& \left\{ x2^\alpha, x2^2, k \cdot x2, a \cdot x2, Fx2^\alpha, FFX2^\alpha, FDx2^\alpha, k \cdot x2, 0, 0, 0, \right. \\
& \quad \left. -a^2 (k \cdot x2), 0, 0, -\frac{m^2 \xi^2 (k \cdot x2)^2}{e^2}, -\frac{m^2 \xi^2 (k \cdot x2)^2}{e^2}, \frac{m^2 \xi^2 (k \cdot x2)^2}{e^2}, 0, 0, 0, 0, 0, 0 \right\} \\
& \left\{ x^\alpha, x^2, k \cdot x, a \cdot x, Fx^\alpha, FFX^\alpha, FDx^\alpha, k \cdot x, 0, 0, 0, -a^2 (k \cdot x), \right. \\
& \quad \left. 0, 0, -\frac{m^2 \xi^2 (k \cdot x)^2}{e^2}, -\frac{m^2 \xi^2 (k \cdot x)^2}{e^2}, \frac{m^2 \xi^2 (k \cdot x)^2}{e^2}, 0, 0, 0, 0, 0, 0 \right\} \\
& \left\{ X^\alpha, X^2, k \cdot X, a \cdot X, FX^\alpha, FFX^\alpha, FDX^\alpha, k \cdot X, 0, 0, 0, \right. \\
& \quad \left. -a^2 (k \cdot X), 0, 0, -\frac{m^2 \xi^2 (k \cdot X)^2}{e^2}, -\frac{m^2 \xi^2 (k \cdot X)^2}{e^2}, \frac{m^2 \xi^2 (k \cdot X)^2}{e^2}, 0, 0, 0, 0, 0, 0 \right\}
\end{aligned}$$

E_p – functions

The functions Ep and EpC are predefined in the file definitions.nb

They provide a list with two data

fields : the preexponent and the phase of the E_p –
function (or the adjoint $\bar{E}_p$ – function)

In[16]:= **Epx1 = Ep[x1, p]**

EqBarx2 = EpC[x2, q]

Out[16]= $\left\{ 1 - \frac{e \phi 1 (\gamma \cdot k) (\gamma \cdot a)}{2 (k \cdot p)}, \frac{a^2 e^2 \phi 1^3}{6 (k \cdot p)} + \frac{e \phi 1^2 (a \cdot p)}{2 (k \cdot p)} - p \cdot x1 \right\}$

Out[17]= $\left\{ 1 - \frac{e \phi 2 (\gamma \cdot a) (\gamma \cdot k)}{2 (k \cdot q)}, -\frac{a^2 e^2 \phi 2^3}{6 (k \cdot q)} - \frac{e \phi 2^2 (a \cdot q)}{2 (k \cdot q)} + q \cdot x2 \right\}$

Propagators in coordinate representation and proper times

s – electron proper time of dimension m^{-2} ;

t – photon proper time of dimension m^{-2}

The functions DiracElectronPropagatorXRepr and

PhotonPropagatorExactXRepr are predefined in the file definitions.nb

They provide a list with three data

fields : a γ -matrix (for S^c) or tensor (for D^c) preexponential,
 a scalar prefactor and the phase of the exponent.
 It is implied that there is an integration over the proper time from 0 to ∞

```
In[18]:= Sc = DiracElectronPropagatorXRepr[x, X, m^2 s];
Sc[[2]] = Simplify[Sc[[2]] * m^2, Assumptions -> {m > 0}];
Sc
```

$$\text{Out[20]} = \left\{ -\frac{e(a \cdot x) \gamma \cdot k}{2m} + \frac{e(\gamma \cdot x)(\gamma \cdot a)(\gamma \cdot k)}{2m} + e s (\gamma \cdot a)(\gamma \cdot k) + \frac{e \phi \gamma \cdot a}{2m} - \frac{1}{3} m \xi^2 s \phi \gamma \cdot k + \frac{\gamma \cdot x}{2ms} + 1, \right. \\ \left. i 2^{-D} e^{-\frac{1}{4} i \pi D} \pi^{-D/2} m \Lambda^{4-D} s^{-D/2}, e \Phi(a \cdot x) - \frac{1}{12} m^2 \xi^2 s \phi^2 - m^2 s - \frac{x^2}{4s} \right\}$$

```
In[21]:= Dc = PhotonPropagatorExactXRepr[x, m^2 t, \mu, \nu];
Dc[[2]] = Simplify[Dc[[2]] * m^2, Assumptions -> {m > 0}];
Dc
```

$$\text{Out[23]} = \left\{ J_2\left(t, \frac{\xi \phi}{2m^2 t}\right) \left(\frac{e^2 (-k^\mu k^\nu (a \cdot x)^2 + a^2 \phi^2 g^{\mu\nu} - a^2 \phi k^\nu x^\mu - a^2 \phi k^\mu x^\nu + \phi a^\nu k^\mu (a \cdot x) + \phi a^\mu k^\nu (a \cdot x) + a^2 x^2 k^\mu k^\nu + \phi^2 (-a^\mu) a^\nu)}{m^2 \xi^2 \phi^2} - \right. \right. \\ \left. \frac{2 i t k^\mu k^\nu}{\phi^2} \right) + J_1\left(t, \frac{\xi \phi}{2m^2 t}\right) \left(\frac{e^2 (k^\mu k^\nu (a \cdot x)^2 - \phi a^\nu k^\mu (a \cdot x) - \phi a^\mu k^\nu (a \cdot x) + \phi^2 a^\mu a^\nu)}{m^2 \xi^2 \phi^2} - \frac{2 i t k^\mu k^\nu}{\phi^2} \right) + \\ \left. g^{\mu\nu} J_0\left(t, \frac{\xi \phi}{2m^2 t}\right), -2^{-D-1} e^{-\frac{1}{4} i \pi D} \pi^{-\frac{D}{2}-1} \Lambda^{4-D} t^{-D/2}, -\frac{x^2}{4t} \right\}$$

Calculation of M

The mass operator

$$M(q, p) = i \int d^D x_1 d^D x_2 \dots;$$

We will write the integrand in the following form

$$\text{Coeff} * \text{Tr}[\text{Matrix}] * \text{Exp}[i \text{Phase}],$$

where

Coeff - is a dimensional coefficient in

front of the expression (also depends on s and t),

Matrix - the γ -matrix factor,

Phase - the total phase of the exponential

We also introduce the notation

$$J_k = J_k\left[t, \frac{\xi \phi}{2m^2 t}\right]$$

```
In[24]:= Coeff = I (I e)^2 Λ^{D-8} Sc[[2]] × Dc[[2]]
Phase = EqBarx2[[2]] + Sc[[3]] + Dc[[3]] + Epx1[[2]]
Matrix =
(EqBarx2[[1]].GAD[μ].Sc[[1]].GAD[ν].Epx1[[1]] × Dc[[1]]) /. {J0[t, ξ φ / 2 / t / m^2] → J0,
J1[t, ξ φ / 2 / t / m^2] → J1, J2[t, ξ φ / 2 / t / m^2] → J2}
```

$$\text{Out[24]} = -2^{-2D-1} e^{-\frac{1}{2} i \pi D} \pi^{-D-1} e^2 m s^{-D/2} t^{-D/2}$$

$$\text{Out[25]} = \frac{a^2 e^2 \phi^3}{6(k \cdot p)} - \frac{a^2 e^2 \phi^2}{6(k \cdot q)} + \frac{e \phi^2 (a \cdot p)}{2(k \cdot p)} - \frac{e \phi^2 (a \cdot q)}{2(k \cdot q)} + e \Phi(a \cdot x) - \frac{1}{12} m^2 \xi^2 s \phi^2 - m^2 s - p \cdot x_1 + q \cdot x_2 - \frac{x^2}{4s} - \frac{x^2}{4t}$$

$$\text{Out[26]} = \left(J_2 \right. \\ \left(\frac{e^2 \left(-k^\mu k^\nu (a \cdot x)^2 + a^2 \phi^2 g^{\mu\nu} - a^2 \phi k^\nu x^\mu - a^2 \phi k^\mu x^\nu + \phi a^\nu k^\mu (a \cdot x) + \phi a^\mu k^\nu (a \cdot x) + a^2 x^2 k^\mu k^\nu + \phi^2 (-a^\mu) a^\nu \right)}{m^2 \xi^2 \phi^2} \right. \\ \left. - \frac{2 i t k^\mu k^\nu}{\phi^2} \right) + J_1 \left(\frac{e^2 \left(k^\mu k^\nu (a \cdot x)^2 - \phi a^\nu k^\mu (a \cdot x) - \phi a^\mu k^\nu (a \cdot x) + \phi^2 a^\mu a^\nu \right)}{m^2 \xi^2 \phi^2} - \frac{2 i t k^\mu k^\nu}{\phi^2} \right) + \\ J_0 g^{\mu\nu} \left(1 - \frac{e \phi^2 (\gamma \cdot a) (\gamma \cdot k)}{2(k \cdot q)} \right) \gamma^\mu. \\ \left(-\frac{e(a \cdot x) \gamma \cdot k}{2m} + \frac{e(\gamma \cdot x) (\gamma \cdot a) (\gamma \cdot k)}{2m} + e s (\gamma \cdot a) (\gamma \cdot k) + \frac{e \phi \gamma \cdot a}{2m} - \frac{1}{3} m \xi^2 s \phi \gamma \cdot k + \frac{\gamma \cdot x}{2m s} + 1 \right). \\ \gamma^\nu \left(1 - \frac{e \phi^2 (\gamma \cdot k) (\gamma \cdot a)}{2(k \cdot p)} \right)$$

γ - matrix algebra

We perform simplifications of the γ

-matrix factor with the aid of FeynCalc functions DotSimplify, DiracSimplify, and then Contract the Lorentz indices.

```
In[27]:= Coeff1 = Coeff;
Phase1 = Phase;
Contract[DotSimplify[Expand[Matrix]]];
Matrix1 = Collect[Contract[DiracSimplify[%]], {J0, J1, J2}]
```

$$\begin{aligned}
\text{Out[30]} = & \text{J0} \left(-\frac{D \phi \phi 1 \gamma \cdot k a^2 e^2}{4 m (k \cdot p)} + \frac{3 \phi \phi 1 \gamma \cdot k a^2 e^2}{2 m (k \cdot p)} + \frac{D \phi \phi 2 \gamma \cdot k a^2 e^2}{4 m (k \cdot q)} - \frac{3 \phi \phi 2 \gamma \cdot k a^2 e^2}{2 m (k \cdot q)} + \frac{D \phi \phi 1 \phi 2 \gamma \cdot k a^2 e^2}{4 m s (k \cdot p) (k \cdot q)} - \right. \\
& \frac{\phi \phi 1 \phi 2 \gamma \cdot k a^2 e^2}{2 m s (k \cdot p) (k \cdot q)} - \frac{D \phi \gamma \cdot a e}{2 m} + \frac{\phi \gamma \cdot a e}{m} + D s (\gamma \cdot a) (\gamma \cdot k) e - 4 s (\gamma \cdot a) (\gamma \cdot k) e - \frac{(\gamma \cdot k) (\gamma \cdot a) (\gamma \cdot x) e}{m} - \\
& \frac{D (\gamma \cdot x) (\gamma \cdot a) (\gamma \cdot k) e}{2 m} + \frac{2 (\gamma \cdot x) (\gamma \cdot a) (\gamma \cdot k) e}{m} + \frac{D \gamma \cdot k (a \cdot x) e}{2 m} - \frac{\gamma \cdot k (a \cdot x) e}{m} - \frac{D \phi 1 (\gamma \cdot k) (\gamma \cdot a) e}{2 (k \cdot p)} + \\
& \frac{D \phi 1 (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) e}{4 m s (k \cdot p)} - \frac{\phi 1 (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) e}{2 m s (k \cdot p)} - \frac{D \phi 2 (\gamma \cdot a) (\gamma \cdot k) e}{2 (k \cdot q)} + \frac{D \phi 2 (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x) e}{4 m s (k \cdot q)} - \\
& \left. \frac{\phi 2 (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x) e}{2 m s (k \cdot q)} + D + \frac{1}{3} D m s \xi^2 \phi \gamma \cdot k - \frac{2}{3} m s \xi^2 \phi \gamma \cdot k - \frac{D \gamma \cdot x}{2 m s} + \frac{\gamma \cdot x}{m s} \right) + \\
& \text{J1} \left(\frac{\phi \phi 1 \gamma \cdot k a^4 e^4}{4 m^3 \xi^2 (k \cdot p)} - \frac{\phi \phi 2 \gamma \cdot k a^4 e^4}{4 m^3 \xi^2 (k \cdot q)} + \frac{\phi \phi 1 \phi 2 \gamma \cdot k a^4 e^4}{4 m^3 s \xi^2 (k \cdot p) (k \cdot q)} + \frac{\phi \gamma \cdot a a^2 e^3}{2 m^3 \xi^2} + \frac{s (\gamma \cdot k) (\gamma \cdot a) a^2 e^3}{m^2 \xi^2} - \right. \\
& \frac{(\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) a^2 e^3}{2 m^3 \xi^2} - \frac{\gamma \cdot k a^2 (a \cdot x) e^3}{2 m^3 \xi^2} - \frac{\phi 1 (\gamma \cdot k) (\gamma \cdot a) a^2 e^3}{2 m^2 \xi^2 (k \cdot p)} + \frac{\phi 1 (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) a^2 e^3}{4 m^3 s \xi^2 (k \cdot p)} - \\
& \frac{\phi 2 (\gamma \cdot a) (\gamma \cdot k) a^2 e^3}{2 m^2 \xi^2 (k \cdot q)} + \frac{\phi 2 (\gamma \cdot k) (\gamma \cdot x) (\gamma \cdot a) a^2 e^3}{4 m^3 s \xi^2 (k \cdot q)} - \frac{\phi 2 \gamma \cdot k a^2 (a \cdot x) e^3}{2 m^3 s \xi^2 (k \cdot q)} + \frac{\gamma \cdot k (a \cdot x)^2 e^2}{m^3 s \xi^2 \phi} + \\
& \frac{s \phi \gamma \cdot k a^2 e^2}{3 m} - \frac{\gamma \cdot x a^2 e^2}{2 m^3 s \xi^2} + \frac{a^2 e^2}{m^2 \xi^2} + \frac{\gamma \cdot a (a \cdot x) e^2}{m^3 s \xi^2} - \frac{(\gamma \cdot a) (\gamma \cdot k) (a \cdot x) e^2}{m^2 \xi^2 \phi} - \\
& \left. \frac{(\gamma \cdot k) (\gamma \cdot a) (a \cdot x) e^2}{m^2 \xi^2 \phi} - \frac{(\gamma \cdot a) (\gamma \cdot x) (\gamma \cdot k) (a \cdot x) e^2}{2 m^3 s \xi^2 \phi} - \frac{(\gamma \cdot k) (\gamma \cdot x) (\gamma \cdot a) (a \cdot x) e^2}{2 m^3 s \xi^2 \phi} - \frac{2 i t \gamma \cdot k}{m s \phi} \right) + \\
& \text{J2} \left(-\frac{D \phi \phi 1 \gamma \cdot k a^4 e^4}{4 m^3 \xi^2 (k \cdot p)} + \frac{3 \phi \phi 1 \gamma \cdot k a^4 e^4}{4 m^3 \xi^2 (k \cdot p)} + \frac{D \phi \phi 2 \gamma \cdot k a^4 e^4}{4 m^3 \xi^2 (k \cdot q)} - \frac{3 \phi \phi 2 \gamma \cdot k a^4 e^4}{4 m^3 \xi^2 (k \cdot q)} + \frac{D \phi \phi 1 \phi 2 \gamma \cdot k a^4 e^4}{4 m^3 s \xi^2 (k \cdot p) (k \cdot q)} - \right. \\
& \frac{3 \phi \phi 1 \phi 2 \gamma \cdot k a^4 e^4}{4 m^3 s \xi^2 (k \cdot p) (k \cdot q)} - \frac{D \phi \gamma \cdot a a^2 e^3}{2 m^3 \xi^2} + \frac{\phi \gamma \cdot a a^2 e^3}{2 m^3 \xi^2} + \frac{D s (\gamma \cdot a) (\gamma \cdot k) a^2 e^3}{m^2 \xi^2} - \frac{4 s (\gamma \cdot a) (\gamma \cdot k) a^2 e^3}{m^2 \xi^2} - \\
& \frac{s (\gamma \cdot k) (\gamma \cdot a) a^2 e^3}{m^2 \xi^2} - \frac{(\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x) a^2 e^3}{m^3 \xi^2} - \frac{3 (\gamma \cdot k) (\gamma \cdot a) (\gamma \cdot x) a^2 e^3}{2 m^3 \xi^2} - \frac{D (\gamma \cdot x) (\gamma \cdot a) (\gamma \cdot k) a^2 e^3}{2 m^3 \xi^2} + \\
& \frac{3 (\gamma \cdot x) (\gamma \cdot a) (\gamma \cdot k) a^2 e^3}{2 m^3 \xi^2} + \frac{(\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) a^2 e^3}{2 m^3 \xi^2} + \frac{D \gamma \cdot k a^2 (a \cdot x) e^3}{2 m^3 \xi^2} - \frac{\gamma \cdot k a^2 (a \cdot x) e^3}{2 m^3 \xi^2} - \\
& \frac{D \phi 1 (\gamma \cdot k) (\gamma \cdot a) a^2 e^3}{2 m^2 \xi^2 (k \cdot p)} + \frac{3 \phi 1 (\gamma \cdot k) (\gamma \cdot a) a^2 e^3}{2 m^2 \xi^2 (k \cdot p)} + \frac{D \phi 1 (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) a^2 e^3}{4 m^3 s \xi^2 (k \cdot p)} - \frac{3 \phi 1 (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) a^2 e^3}{4 m^3 s \xi^2 (k \cdot p)} - \\
& \frac{D \phi 2 (\gamma \cdot a) (\gamma \cdot k) a^2 e^3}{2 m^2 \xi^2 (k \cdot q)} + \frac{3 \phi 2 (\gamma \cdot a) (\gamma \cdot k) a^2 e^3}{2 m^2 \xi^2 (k \cdot q)} + \frac{D \phi 2 (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x) a^2 e^3}{4 m^3 s \xi^2 (k \cdot q)} - \frac{\phi 2 (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x) a^2 e^3}{2 m^3 s \xi^2 (k \cdot q)} - \\
& \frac{\phi 2 (\gamma \cdot k) (\gamma \cdot x) (\gamma \cdot a) a^2 e^3}{4 m^3 s \xi^2 (k \cdot q)} + \frac{\phi 2 \gamma \cdot k a^2 (a \cdot x) e^3}{2 m^3 s \xi^2 (k \cdot q)} - \frac{\gamma \cdot k (a \cdot x)^2 e^2}{m^3 s \xi^2 \phi} + \frac{D s \phi \gamma \cdot k a^2 e^2}{3 m} - \frac{s \phi \gamma \cdot k a^2 e^2}{m} - \\
& \frac{D \gamma \cdot x a^2 e^2}{2 m^3 s \xi^2} + \frac{3 \gamma \cdot x a^2 e^2}{2 m^3 s \xi^2} - \frac{(\gamma \cdot k) (\gamma \cdot x) a^2 e^2}{m^2 \xi^2 \phi} - \frac{(\gamma \cdot x) (\gamma \cdot k) a^2 e^2}{m^2 \xi^2 \phi} + \frac{D a^2 e^2}{m^2 \xi^2} - \frac{a^2 e^2}{m^2 \xi^2} - \frac{\gamma \cdot a (a \cdot x) e^2}{m^3 s \xi^2} + \\
& \left. \frac{(\gamma \cdot a) (\gamma \cdot k) (a \cdot x) e^2}{m^2 \xi^2 \phi} + \frac{(\gamma \cdot k) (\gamma \cdot a) (a \cdot x) e^2}{m^2 \xi^2 \phi} + \frac{(\gamma \cdot a) (\gamma \cdot x) (\gamma \cdot k) (a \cdot x) e^2}{2 m^3 s \xi^2 \phi} + \frac{(\gamma \cdot k) (\gamma \cdot x) (\gamma \cdot a) (a \cdot x) e^2}{2 m^3 s \xi^2 \phi} - \frac{2 i t \gamma \cdot k}{m s \phi} \right)
\end{aligned}$$

Substitution of the variables

$$x1 = x - \frac{x}{2};$$

$$x_2 = x + \frac{X}{2};$$

$$\phi_1 = \phi - \frac{\Phi}{2};$$

$$\phi_2 = \phi + \frac{\Phi}{2};$$

$$\phi = kx = m x_-;$$

$$\Phi = kX = m X_-;$$

The integration measure :

$$d^D x_1 d^D x_2 = d^D x d^D X$$

In[31]:= **Coeff2 = Coeff1;**

Phase2 = Expand[ExpandScalarProduct[

Phase1 /. {φ2 → Φ + φ / 2, φ1 → Φ - φ / 2, av2 → -m^2 ξ^2 / e^2} /.

{Momentum[x1, D] → Momentum[X, D] - Momentum[x, D] / 2,

Momentum[x2, D] → Momentum[X, D] + Momentum[x, D] / 2}

]]

Matrix2 = Collect[

Expand[

Matrix1 /. {φ2 → Φ + φ / 2, φ1 → Φ - φ / 2, av2 → -m^2 ξ^2 / e^2}

],

{J0, J1, J2}]

$$\begin{aligned} \text{Out[32]} = & \frac{e \Phi^2 (a \cdot p)}{2 (k \cdot p)} + \frac{e \phi^2 (a \cdot p)}{8 (k \cdot p)} - \frac{e \Phi \phi (a \cdot p)}{2 (k \cdot p)} - \frac{e \Phi^2 (a \cdot q)}{2 (k \cdot q)} - \frac{e \phi^2 (a \cdot q)}{8 (k \cdot q)} - \frac{e \Phi \phi (a \cdot q)}{2 (k \cdot q)} + \\ & e \Phi (a \cdot x) - \frac{m^2 \xi^2 \Phi^3}{6 (k \cdot p)} + \frac{m^2 \xi^2 \phi^3}{48 (k \cdot p)} - \frac{m^2 \xi^2 \Phi \phi^2}{8 (k \cdot p)} + \frac{m^2 \xi^2 \Phi^2 \phi}{4 (k \cdot p)} + \frac{m^2 \xi^2 \Phi^3}{6 (k \cdot q)} + \frac{m^2 \xi^2 \phi^3}{48 (k \cdot q)} \\ & - \frac{m^2 \xi^2 \Phi \phi^2}{8 (k \cdot q)} + \frac{m^2 \xi^2 \Phi^2 \phi}{4 (k \cdot q)} - \frac{1}{12} m^2 \xi^2 s \phi^2 - m^2 s + \frac{p \cdot x}{2} - p \cdot X + \frac{q \cdot x}{2} + q \cdot X - \frac{x^2}{4 s} - \frac{x^2}{4 t} \end{aligned}$$

$$\begin{aligned} \text{Out[33]} = & J1 \left(-\frac{m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p) (k \cdot q)} - \frac{m \xi^2 \gamma \cdot k \phi^2}{8 (k \cdot p)} - \frac{m \xi^2 \gamma \cdot k \phi^2}{8 (k \cdot q)} - \frac{e \gamma \cdot a \phi}{2 m} - \frac{1}{3} m s \xi^2 \gamma \cdot k \phi + \right. \\ & \frac{m \xi^2 \Phi \gamma \cdot k \phi}{4 (k \cdot p)} - \frac{e (\gamma \cdot k) \cdot (\gamma \cdot a) \phi}{4 (k \cdot p)} + \frac{e (\gamma \cdot x) \cdot (\gamma \cdot k) \cdot (\gamma \cdot a) \phi}{8 m s (k \cdot p)} - \frac{m \xi^2 \Phi \gamma \cdot k \phi}{4 (k \cdot q)} + \frac{e (\gamma \cdot a) \cdot (\gamma \cdot k) \phi}{4 (k \cdot q)} - \\ & \frac{e (\gamma \cdot k) \cdot (\gamma \cdot x) \cdot (\gamma \cdot a) \phi}{8 m s (k \cdot q)} + \frac{e \gamma \cdot k (a \cdot x) \phi}{4 m s (k \cdot q)} + \frac{m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p) (k \cdot q)} + \frac{\gamma \cdot x}{2 m s} - e s (\gamma \cdot k) \cdot (\gamma \cdot a) + \\ & \frac{e (\gamma \cdot x) \cdot (\gamma \cdot k) \cdot (\gamma \cdot a)}{2 m} + \frac{e^2 \gamma \cdot a (a \cdot x)}{m^3 s \xi^2} + \frac{e \gamma \cdot k (a \cdot x)}{2 m} + \frac{e \Phi (\gamma \cdot k) \cdot (\gamma \cdot a)}{2 (k \cdot p)} - \frac{e \Phi (\gamma \cdot x) \cdot (\gamma \cdot k) \cdot (\gamma \cdot a)}{4 m s (k \cdot p)} + \\ & \frac{e \Phi (\gamma \cdot a) \cdot (\gamma \cdot k)}{2 (k \cdot q)} - \frac{e \Phi (\gamma \cdot k) \cdot (\gamma \cdot x) \cdot (\gamma \cdot a)}{4 m s (k \cdot q)} + \frac{e \Phi \gamma \cdot k (a \cdot x)}{2 m s (k \cdot q)} - 1 + \frac{e^2 \gamma \cdot k (a \cdot x)^2}{m^3 s \xi^2 \phi} - \frac{2 i t \gamma \cdot k}{m s \phi} \end{aligned}$$

$$\begin{aligned}
& \left. \frac{e^2 (\gamma \cdot a) (\gamma \cdot k) (a \cdot x)}{m^2 \xi^2 \phi} - \frac{e^2 (\gamma \cdot k) (\gamma \cdot a) (a \cdot x)}{m^2 \xi^2 \phi} - \frac{e^2 (\gamma \cdot a) (\gamma \cdot x) (\gamma \cdot k) (a \cdot x)}{2 m^3 s \xi^2 \phi} - \frac{e^2 (\gamma \cdot k) (\gamma \cdot x) (\gamma \cdot a) (a \cdot x)}{2 m^3 s \xi^2 \phi} \right) + \\
& \text{J0} \left(\frac{D m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p) (k \cdot q)} - \frac{m \xi^2 \gamma \cdot k \phi^3}{8 s (k \cdot p) (k \cdot q)} - \frac{D m \xi^2 \gamma \cdot k \phi^2}{8 (k \cdot p)} + \frac{3 m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} - \frac{D m \xi^2 \gamma \cdot k \phi^2}{8 (k \cdot q)} + \frac{3 m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot q)} - \right. \\
& \quad \frac{D e \gamma \cdot a \phi}{2 m} + \frac{e \gamma \cdot a \phi}{m} + \frac{1}{3} D m s \xi^2 \gamma \cdot k \phi - \frac{2}{3} m s \xi^2 \gamma \cdot k \phi + \frac{D m \xi^2 \Phi \gamma \cdot k \phi}{4 (k \cdot p)} - \frac{3 m \xi^2 \Phi \gamma \cdot k \phi}{2 (k \cdot p)} + \\
& \quad \frac{D e (\gamma \cdot k) (\gamma \cdot a) \phi}{4 (k \cdot p)} - \frac{D e (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) \phi}{8 m s (k \cdot p)} + \frac{e (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) \phi}{4 m s (k \cdot p)} - \frac{D m \xi^2 \Phi \gamma \cdot k \phi}{4 (k \cdot q)} + \frac{3 m \xi^2 \Phi \gamma \cdot k \phi}{2 (k \cdot q)} - \\
& \quad \frac{D e (\gamma \cdot a) (\gamma \cdot k) \phi}{4 (k \cdot q)} + \frac{D e (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x) \phi}{8 m s (k \cdot q)} - \frac{e (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x) \phi}{4 m s (k \cdot q)} - \frac{D m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p) (k \cdot q)} + \frac{m \xi^2 \Phi^2 \gamma \cdot k \phi}{2 s (k \cdot p) (k \cdot q)} + \\
& \quad D - \frac{D \gamma \cdot x}{2 m s} + \frac{\gamma \cdot x}{m s} + D e s (\gamma \cdot a) (\gamma \cdot k) - 4 e s (\gamma \cdot a) (\gamma \cdot k) - \frac{e (\gamma \cdot k) (\gamma \cdot a) (\gamma \cdot x)}{m} - \frac{D e (\gamma \cdot x) (\gamma \cdot a) (\gamma \cdot k)}{2 m} + \\
& \quad \frac{2 e (\gamma \cdot x) (\gamma \cdot a) (\gamma \cdot k)}{m} + \frac{D e \gamma \cdot k (a \cdot x)}{2 m} - \frac{e \gamma \cdot k (a \cdot x)}{m} - \frac{D e \Phi (\gamma \cdot k) (\gamma \cdot a)}{2 (k \cdot p)} + \frac{D e \Phi (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a)}{4 m s (k \cdot p)} - \\
& \quad \left. \frac{e \Phi (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a)}{2 m s (k \cdot p)} - \frac{D e \Phi (\gamma \cdot a) (\gamma \cdot k)}{2 (k \cdot q)} + \frac{D e \Phi (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x)}{4 m s (k \cdot q)} - \frac{e \Phi (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x)}{2 m s (k \cdot q)} \right) + \\
& \text{J2} \left(- \frac{D m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p) (k \cdot q)} + \frac{3 m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p) (k \cdot q)} + \frac{D m \xi^2 \gamma \cdot k \phi^2}{8 (k \cdot p)} - \frac{3 m \xi^2 \gamma \cdot k \phi^2}{8 (k \cdot p)} + \frac{D m \xi^2 \gamma \cdot k \phi^2}{8 (k \cdot q)} - \frac{3 m \xi^2 \gamma \cdot k \phi^2}{8 (k \cdot q)} + \right. \\
& \quad \frac{D e \gamma \cdot a \phi}{2 m} - \frac{e \gamma \cdot a \phi}{2 m} - \frac{1}{3} D m s \xi^2 \gamma \cdot k \phi + m s \xi^2 \gamma \cdot k \phi - \frac{D m \xi^2 \Phi \gamma \cdot k \phi}{4 (k \cdot p)} + \frac{3 m \xi^2 \Phi \gamma \cdot k \phi}{4 (k \cdot p)} - \\
& \quad \frac{D e (\gamma \cdot k) (\gamma \cdot a) \phi}{4 (k \cdot p)} + \frac{3 e (\gamma \cdot k) (\gamma \cdot a) \phi}{4 (k \cdot p)} + \frac{D e (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) \phi}{8 m s (k \cdot p)} - \frac{3 e (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) \phi}{8 m s (k \cdot p)} + \frac{D m \xi^2 \Phi \gamma \cdot k \phi}{4 (k \cdot q)} - \\
& \quad \frac{3 m \xi^2 \Phi \gamma \cdot k \phi}{4 (k \cdot q)} + \frac{D e (\gamma \cdot a) (\gamma \cdot k) \phi}{4 (k \cdot q)} - \frac{3 e (\gamma \cdot a) (\gamma \cdot k) \phi}{4 (k \cdot q)} - \frac{D e (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x) \phi}{8 m s (k \cdot q)} + \frac{e (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x) \phi}{4 m s (k \cdot q)} + \\
& \quad \frac{e (\gamma \cdot k) (\gamma \cdot x) (\gamma \cdot a) \phi}{8 m s (k \cdot q)} - \frac{e \gamma \cdot k (a \cdot x) \phi}{4 m s (k \cdot q)} + \frac{D m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p) (k \cdot q)} - \frac{3 m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p) (k \cdot q)} - D + \frac{D \gamma \cdot x}{2 m s} - \frac{3 \gamma \cdot x}{2 m s} - \\
& \quad D e s (\gamma \cdot a) (\gamma \cdot k) + 4 e s (\gamma \cdot a) (\gamma \cdot k) + e s (\gamma \cdot k) (\gamma \cdot a) + \frac{e (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x)}{m} + \frac{3 e (\gamma \cdot k) (\gamma \cdot a) (\gamma \cdot x)}{2 m} + \\
& \quad \frac{D e (\gamma \cdot x) (\gamma \cdot a) (\gamma \cdot k)}{2 m} - \frac{3 e (\gamma \cdot x) (\gamma \cdot a) (\gamma \cdot k)}{2 m} - \frac{e (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a)}{2 m} - \frac{e^2 \gamma \cdot a (a \cdot x)}{m^3 s \xi^2} - \frac{D e \gamma \cdot k (a \cdot x)}{2 m} + \\
& \quad \frac{e \gamma \cdot k (a \cdot x)}{2 m} + \frac{D e \Phi (\gamma \cdot k) (\gamma \cdot a)}{2 (k \cdot p)} - \frac{3 e \Phi (\gamma \cdot k) (\gamma \cdot a)}{2 (k \cdot p)} - \frac{D e \Phi (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a)}{4 m s (k \cdot p)} + \frac{3 e \Phi (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a)}{4 m s (k \cdot p)} + \\
& \quad \frac{D e \Phi (\gamma \cdot a) (\gamma \cdot k)}{2 (k \cdot q)} - \frac{3 e \Phi (\gamma \cdot a) (\gamma \cdot k)}{2 (k \cdot q)} - \frac{D e \Phi (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x)}{4 m s (k \cdot q)} + \frac{e \Phi (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x)}{2 m s (k \cdot q)} + \\
& \quad \frac{e \Phi (\gamma \cdot k) (\gamma \cdot x) (\gamma \cdot a)}{4 m s (k \cdot q)} - \frac{e \Phi \gamma \cdot k (a \cdot x)}{2 m s (k \cdot q)} + 1 - \frac{e^2 \gamma \cdot k (a \cdot x)^2}{m^3 s \xi^2 \phi} - \frac{2 i t \gamma \cdot k}{m s \phi} + \frac{(\gamma \cdot k) (\gamma \cdot x)}{\phi} + \frac{(\gamma \cdot x) (\gamma \cdot k)}{\phi} + \\
& \quad \left. \frac{e^2 (\gamma \cdot a) (\gamma \cdot k) (a \cdot x)}{m^2 \xi^2 \phi} + \frac{e^2 (\gamma \cdot k) (\gamma \cdot a) (a \cdot x)}{m^2 \xi^2 \phi} + \frac{e^2 (\gamma \cdot a) (\gamma \cdot x) (\gamma \cdot k) (a \cdot x)}{2 m^3 s \xi^2 \phi} + \frac{e^2 (\gamma \cdot k) (\gamma \cdot x) (\gamma \cdot a) (a \cdot x)}{2 m^3 s \xi^2 \phi} \right)
\end{aligned}$$

Integration over

$$\int d^{D-2} X_{\perp} dX_{+} \dots$$

Preexp does not depend on $X_{\perp}$ and

X_{+} . Phase contains $X_{\perp}$ and X_{+} in scalar products

$$pX = p_{-} X_{+} + p_{+} X_{-} - p_{\perp} X_{\perp};$$

$$qX = q_{-} X_{+} + q_{+} X_{-} - q_{\perp} X_{\perp};$$

so

$$\int d^{D-2} X_{\perp} dX_{+} e^{-i(p-q) \cdot X} \dots = (2\pi)^{D-1} \delta^{(D-2)}(p_{\perp} - q_{\perp}) \delta(p_{-} - q_{-}).$$

We will not write the δ - functions explicitly,
but we will assume that they are present;

Due to the conservation law $\delta^{(D-2)}(p_{\perp} - q_{\perp}) \delta(p_{-} - q_{-})$

$$kq \rightarrow kp$$

$$aq \rightarrow ap$$

Notations

$$pp = p_{+};$$

$$qp = q_{+};$$

The remaining integrals : $\int dX_{\perp} d^D x \dots$

In[34]:= **Coeff3 = Coeff2 * (2 π) ^ (D - 1)**

Phase3 =

**Phase2 /. {Pair[Momentum[p, D], Momentum[X, D]] $\rightarrow$ pp Xm, Pair[Momentum[q, D],
Momentum[X, D]] $\rightarrow$ qp Xm} /. {Xm $\rightarrow$ Φ / m} /. {kq $\rightarrow$ kp, aq $\rightarrow$ ap}**

Matrix3 = Matrix2 /. {kq $\rightarrow$ kp, aq $\rightarrow$ ap}

$$\text{Out[34]} = -\frac{2^{-D-2} e^{-\frac{1}{2} i \pi D} e^2 m s^{-D/2} t^{-D/2}}{\pi^2}$$

$$\text{Out[35]} = -\frac{e \Phi \phi(a \cdot p)}{k \cdot p} + e \Phi(a \cdot x) + \frac{m^2 \xi^2 \phi^3}{24 (k \cdot p)} + \frac{m^2 \xi^2 \Phi^2 \phi}{2 (k \cdot p)} - \frac{1}{12} m^2 \xi^2 s \phi^2 - m^2 s - \frac{pp \Phi}{m} + \frac{qp \Phi}{m} + \frac{p \cdot x}{2} + \frac{q \cdot x}{2} - \frac{x^2}{4 s} - \frac{x^2}{4 t}$$

$$\begin{aligned} \text{Out[36]} = & J0 \left(\frac{D m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} - \frac{m \xi^2 \gamma \cdot k \phi^3}{8 s (k \cdot p)^2} - \frac{D m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} + \frac{3 m \xi^2 \gamma \cdot k \phi^2}{2 (k \cdot p)} - \frac{D e \gamma \cdot a \phi}{2 m} + \frac{e \gamma \cdot a \phi}{m} + \frac{1}{3} D m s \xi^2 \gamma \cdot k \phi - \right. \\ & \frac{2}{3} m s \xi^2 \gamma \cdot k \phi - \frac{D e (\gamma \cdot a) (\gamma \cdot k) \phi}{4 (k \cdot p)} + \frac{D e (\gamma \cdot k) (\gamma \cdot a) \phi}{4 (k \cdot p)} + \frac{D e (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x) \phi}{8 m s (k \cdot p)} - \frac{e (\gamma \cdot a) (\gamma \cdot k) (\gamma \cdot x) \phi}{4 m s (k \cdot p)} \\ & \frac{D e (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) \phi}{8 m s (k \cdot p)} + \frac{e (\gamma \cdot x) (\gamma \cdot k) (\gamma \cdot a) \phi}{4 m s (k \cdot p)} - \frac{D m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} + \frac{m \xi^2 \Phi^2 \gamma \cdot k \phi}{2 s (k \cdot p)^2} + D - \frac{D \gamma \cdot x}{2 m s} \\ & \left. \frac{\gamma \cdot x}{m s} + D e s (\gamma \cdot a) (\gamma \cdot k) - 4 e s (\gamma \cdot a) (\gamma \cdot k) - \frac{e (\gamma \cdot k) (\gamma \cdot a) (\gamma \cdot x)}{m} - \frac{D e (\gamma \cdot x) (\gamma \cdot a) (\gamma \cdot k)}{2 m} + \right) \end{aligned}$$

$$\begin{aligned}
& \frac{2e(\gamma \cdot x)(\gamma \cdot a)(\gamma \cdot k)}{m} + \frac{De\gamma \cdot k(a \cdot x)}{2m} - \frac{e\gamma \cdot k(a \cdot x)}{m} - \frac{De\Phi(\gamma \cdot a)(\gamma \cdot k)}{2(k \cdot p)} - \frac{De\Phi(\gamma \cdot k)(\gamma \cdot a)}{2(k \cdot p)} + \\
& \left. \frac{De\Phi(\gamma \cdot a)(\gamma \cdot k)(\gamma \cdot x)}{4ms(k \cdot p)} - \frac{e\Phi(\gamma \cdot a)(\gamma \cdot k)(\gamma \cdot x)}{2ms(k \cdot p)} + \frac{De\Phi(\gamma \cdot x)(\gamma \cdot k)(\gamma \cdot a)}{4ms(k \cdot p)} - \frac{e\Phi(\gamma \cdot x)(\gamma \cdot k)(\gamma \cdot a)}{2ms(k \cdot p)} \right) + \\
J2 & \left(-\frac{Dm\xi^2\gamma \cdot k\phi^3}{16s(k \cdot p)^2} + \frac{3m\xi^2\gamma \cdot k\phi^3}{16s(k \cdot p)^2} + \frac{Dm\xi^2\gamma \cdot k\phi^2}{4(k \cdot p)} - \frac{3m\xi^2\gamma \cdot k\phi^2}{4(k \cdot p)} + \frac{De\gamma \cdot a\phi}{2m} - \frac{e\gamma \cdot a\phi}{2m} - \frac{1}{3}Dms\xi^2\gamma \cdot k\phi + \right. \\
& m s \xi^2 \gamma \cdot k \phi + \frac{De(\gamma \cdot a)(\gamma \cdot k)\phi}{4(k \cdot p)} - \frac{3e(\gamma \cdot a)(\gamma \cdot k)\phi}{4(k \cdot p)} - \frac{De(\gamma \cdot k)(\gamma \cdot a)\phi}{4(k \cdot p)} + \frac{3e(\gamma \cdot k)(\gamma \cdot a)\phi}{4(k \cdot p)} - \\
& \frac{De(\gamma \cdot a)(\gamma \cdot k)(\gamma \cdot x)\phi}{8ms(k \cdot p)} + \frac{e(\gamma \cdot a)(\gamma \cdot k)(\gamma \cdot x)\phi}{4ms(k \cdot p)} + \frac{e(\gamma \cdot k)(\gamma \cdot x)(\gamma \cdot a)\phi}{8ms(k \cdot p)} + \frac{De(\gamma \cdot x)(\gamma \cdot k)(\gamma \cdot a)\phi}{8ms(k \cdot p)} - \\
& \frac{3e(\gamma \cdot x)(\gamma \cdot k)(\gamma \cdot a)\phi}{8ms(k \cdot p)} - \frac{e\gamma \cdot k(a \cdot x)\phi}{4ms(k \cdot p)} + \frac{Dm\xi^2\Phi^2\gamma \cdot k\phi}{4s(k \cdot p)^2} - \frac{3m\xi^2\Phi^2\gamma \cdot k\phi}{4s(k \cdot p)^2} - D + \frac{D\gamma \cdot x}{2ms} - \frac{3\gamma \cdot x}{2ms} - \\
& Des(\gamma \cdot a)(\gamma \cdot k) + 4es(\gamma \cdot a)(\gamma \cdot k) + es(\gamma \cdot k)(\gamma \cdot a) + \frac{e(\gamma \cdot a)(\gamma \cdot k)(\gamma \cdot x)}{m} + \frac{3e(\gamma \cdot k)(\gamma \cdot a)(\gamma \cdot x)}{2m} + \\
& \frac{De(\gamma \cdot x)(\gamma \cdot a)(\gamma \cdot k)}{2m} - \frac{3e(\gamma \cdot x)(\gamma \cdot a)(\gamma \cdot k)}{2m} - \frac{e(\gamma \cdot x)(\gamma \cdot k)(\gamma \cdot a)}{2m} - \frac{e^2\gamma \cdot a(a \cdot x)}{m^3s\xi^2} - \frac{De\gamma \cdot k(a \cdot x)}{2m} + \\
& \frac{e\gamma \cdot k(a \cdot x)}{2m} + \frac{De\Phi(\gamma \cdot a)(\gamma \cdot k)}{2(k \cdot p)} - \frac{3e\Phi(\gamma \cdot a)(\gamma \cdot k)}{2(k \cdot p)} + \frac{De\Phi(\gamma \cdot k)(\gamma \cdot a)}{2(k \cdot p)} - \frac{3e\Phi(\gamma \cdot k)(\gamma \cdot a)}{2(k \cdot p)} - \\
& \frac{De\Phi(\gamma \cdot a)(\gamma \cdot k)(\gamma \cdot x)}{4ms(k \cdot p)} + \frac{e\Phi(\gamma \cdot a)(\gamma \cdot k)(\gamma \cdot x)}{2ms(k \cdot p)} + \frac{e\Phi(\gamma \cdot k)(\gamma \cdot x)(\gamma \cdot a)}{4ms(k \cdot p)} - \frac{De\Phi(\gamma \cdot x)(\gamma \cdot k)(\gamma \cdot a)}{4ms(k \cdot p)} + \\
& \frac{3e\Phi(\gamma \cdot x)(\gamma \cdot k)(\gamma \cdot a)}{4ms(k \cdot p)} - \frac{e\Phi\gamma \cdot k(a \cdot x)}{2ms(k \cdot p)} + 1 - \frac{e^2\gamma \cdot k(a \cdot x)^2}{m^3s\xi^2\phi} - \frac{2it\gamma \cdot k}{ms\phi} + \frac{(\gamma \cdot k)(\gamma \cdot x)}{\phi} + \frac{(\gamma \cdot x)(\gamma \cdot k)}{\phi} + \\
& \left. \frac{e^2(\gamma \cdot a)(\gamma \cdot k)(a \cdot x)}{m^2\xi^2\phi} + \frac{e^2(\gamma \cdot k)(\gamma \cdot a)(a \cdot x)}{m^2\xi^2\phi} + \frac{e^2(\gamma \cdot a)(\gamma \cdot x)(\gamma \cdot k)(a \cdot x)}{2m^3s\xi^2\phi} + \frac{e^2(\gamma \cdot k)(\gamma \cdot x)(\gamma \cdot a)(a \cdot x)}{2m^3s\xi^2\phi} \right) + \\
J1 & \left(-\frac{m\xi^2\gamma \cdot k\phi^3}{16s(k \cdot p)^2} - \frac{m\xi^2\gamma \cdot k\phi^2}{4(k \cdot p)} - \frac{e\gamma \cdot a\phi}{2m} - \frac{1}{3}ms\xi^2\gamma \cdot k\phi + \frac{e(\gamma \cdot a)(\gamma \cdot k)\phi}{4(k \cdot p)} - \frac{e(\gamma \cdot k)(\gamma \cdot a)\phi}{4(k \cdot p)} - \right. \\
& \frac{e(\gamma \cdot k)(\gamma \cdot x)(\gamma \cdot a)\phi}{8ms(k \cdot p)} + \frac{e(\gamma \cdot x)(\gamma \cdot k)(\gamma \cdot a)\phi}{8ms(k \cdot p)} + \frac{e\gamma \cdot k(a \cdot x)\phi}{4ms(k \cdot p)} + \frac{m\xi^2\Phi^2\gamma \cdot k\phi}{4s(k \cdot p)^2} + \frac{\gamma \cdot x}{2ms} - \\
& es(\gamma \cdot k)(\gamma \cdot a) + \frac{e(\gamma \cdot x)(\gamma \cdot k)(\gamma \cdot a)}{2m} + \frac{e^2\gamma \cdot a(a \cdot x)}{m^3s\xi^2} + \frac{e\gamma \cdot k(a \cdot x)}{2m} + \frac{e\Phi(\gamma \cdot a)(\gamma \cdot k)}{2(k \cdot p)} + \frac{e\Phi(\gamma \cdot k)(\gamma \cdot a)}{2(k \cdot p)} - \\
& \frac{e\Phi(\gamma \cdot k)(\gamma \cdot x)(\gamma \cdot a)}{4ms(k \cdot p)} - \frac{e\Phi(\gamma \cdot x)(\gamma \cdot k)(\gamma \cdot a)}{4ms(k \cdot p)} + \frac{e\Phi\gamma \cdot k(a \cdot x)}{2ms(k \cdot p)} - 1 + \frac{e^2\gamma \cdot k(a \cdot x)^2}{m^3s\xi^2\phi} - \frac{2it\gamma \cdot k}{ms\phi} - \\
& \left. \frac{e^2(\gamma \cdot a)(\gamma \cdot k)(a \cdot x)}{m^2\xi^2\phi} - \frac{e^2(\gamma \cdot k)(\gamma \cdot a)(a \cdot x)}{m^2\xi^2\phi} - \frac{e^2(\gamma \cdot a)(\gamma \cdot x)(\gamma \cdot k)(a \cdot x)}{2m^3s\xi^2\phi} - \frac{e^2(\gamma \cdot k)(\gamma \cdot x)(\gamma \cdot a)(a \cdot x)}{2m^3s\xi^2\phi} \right)
\end{aligned}$$

Reordering the γ - matrix terms in the preexponent

We employ the equality

$$(\gamma a) (\gamma b) (\gamma c) \rightarrow -i \gamma^\beta \cdot \overline{\gamma}^5 \epsilon^{\beta \mu \nu \delta} a_\mu b_\nu c_\delta + (a b) (\gamma c) - (a c) (\gamma b) + (\gamma a) (b c)$$

to rewrite the terms with 3 gamma matrices

Then we recollect tensors $F_{\mu\nu}$,

$F^*_{\mu\nu}$ and $(F^2)_{\mu\nu}$ from the combinations of a_μ , k_μ and the antisymmetric tensor $\epsilon^{\alpha\beta\mu\nu}$

The scalar products (γa) and (γk) can be expressed as

$$(\gamma a) = \frac{1}{\phi} [(\gamma k) (ax) - (\gamma Fx)]$$

$$(\gamma k) = (\gamma k) (kx) / \phi = -a^2 k_\mu k_\nu \gamma^\mu x^\nu \frac{1}{-a^2 \phi} = \frac{e^2}{m^2 \xi^2 \phi} (\gamma F^2 x)$$

Then the result can be expressed as a linear

combination of following γ - matrix structures :

1,

(γx) ,

$(\gamma F^2 x)$,

$(\sigma F) = \sigma_{\mu\nu} F^{\mu\nu}$,

$\gamma^\beta \gamma^5 (F^* x)_\beta$.

Also note that we treat $\overline{\gamma}^5$ as a 4 - dimensional object,

assuming that the terms incorporating it are finite

```
In[37]:= Coeff4 = Coeff3;
Phase4 = Phase3;
Matrix4 = Collect[
  (((Expand[Matrix3 /. TripleGamma]) /. EpsToF) /. FieldSubstitutions) /. akToF /.
  {DiracGamma[Momentum[k, D], D].DiracGamma[Momentum[x, D], D] / \phi \rightarrow
  -DiracGamma[Momentum[x, D], D].DiracGamma[Momentum[k, D], D] / \phi +
  2}, {J0, J1, J2}]
```

$$\begin{aligned}
\text{Out[39]= } & \text{J1} \left(-\frac{m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} - \frac{m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} - \frac{1}{3} m s \xi^2 \gamma \cdot k \phi + \frac{i e \sigma F \phi}{4 (k \cdot p)} - \frac{e \Phi \gamma \cdot a \phi}{2 m s (k \cdot p)} - \frac{i e \gamma^\beta \bar{\gamma}^5 F D x^\beta \phi}{4 m s (k \cdot p)} + \right. \\
& \left. \frac{m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} + \frac{1}{2} i e s \sigma F + \frac{\gamma \cdot x}{2 m s} - \frac{i e \gamma^\beta \bar{\gamma}^5 F D x^\beta}{2 m} + \frac{e \Phi \gamma \cdot k (a \cdot x)}{2 m s (k \cdot p)} - 1 - \frac{2 i t \gamma \cdot k}{m s \phi} \right) + \\
& \text{J0} \left(\frac{D m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} - \frac{m \xi^2 \gamma \cdot k \phi^3}{8 s (k \cdot p)^2} - \frac{D m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} + \frac{3 m \xi^2 \gamma \cdot k \phi^2}{2 (k \cdot p)} + \frac{1}{3} D m s \xi^2 \gamma \cdot k \phi - \right. \\
& \frac{2}{3} m s \xi^2 \gamma \cdot k \phi - \frac{i D e \sigma F \phi}{4 (k \cdot p)} + \frac{D e \Phi \gamma \cdot a \phi}{2 m s (k \cdot p)} - \frac{e \Phi \gamma \cdot a \phi}{m s (k \cdot p)} + \frac{i D e \gamma^\beta \bar{\gamma}^5 F D x^\beta \phi}{4 m s (k \cdot p)} - \\
& \frac{i e \gamma^\beta \bar{\gamma}^5 F D x^\beta \phi}{2 m s (k \cdot p)} - \frac{D m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} + \frac{m \xi^2 \Phi^2 \gamma \cdot k \phi}{2 s (k \cdot p)^2} + D + \frac{1}{2} i D e s \sigma F - 2 i e s \sigma F - \\
& \left. \frac{D \gamma \cdot x}{2 m s} + \frac{\gamma \cdot x}{m s} - \frac{i D e \gamma^\beta \bar{\gamma}^5 F D x^\beta}{2 m} + \frac{3 i e \gamma^\beta \bar{\gamma}^5 F D x^\beta}{m} - \frac{D e \Phi \gamma \cdot k (a \cdot x)}{2 m s (k \cdot p)} + \frac{e \Phi \gamma \cdot k (a \cdot x)}{m s (k \cdot p)} \right) + \\
& \text{J2} \left(-\frac{D m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} + \frac{3 m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} + \frac{D m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} - \frac{3 m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} - \frac{1}{3} D m s \xi^2 \gamma \cdot k \phi + m s \xi^2 \gamma \cdot k \phi + \right. \\
& \frac{i D e \sigma F \phi}{4 (k \cdot p)} - \frac{3 i e \sigma F \phi}{4 (k \cdot p)} - \frac{D e \Phi \gamma \cdot a \phi}{2 m s (k \cdot p)} + \frac{3 e \Phi \gamma \cdot a \phi}{2 m s (k \cdot p)} - \frac{i D e \gamma^\beta \bar{\gamma}^5 F D x^\beta \phi}{4 m s (k \cdot p)} + \frac{3 i e \gamma^\beta \bar{\gamma}^5 F D x^\beta \phi}{4 m s (k \cdot p)} + \\
& \frac{D m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} - \frac{3 m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} - D - \frac{1}{2} i D e s \sigma F + \frac{3}{2} i e s \sigma F + \frac{D \gamma \cdot x}{2 m s} - \frac{3 \gamma \cdot x}{2 m s} + \\
& \left. \frac{i D e \gamma^\beta \bar{\gamma}^5 F D x^\beta}{2 m} - \frac{3 i e \gamma^\beta \bar{\gamma}^5 F D x^\beta}{2 m} + \frac{D e \Phi \gamma \cdot k (a \cdot x)}{2 m s (k \cdot p)} - \frac{3 e \Phi \gamma \cdot k (a \cdot x)}{2 m s (k \cdot p)} + 3 - \frac{2 i t \gamma \cdot k}{m s \phi} \right)
\end{aligned}$$

Substituting the lightcone variables

We introduce the following notations

$$\begin{aligned}(xp, pp, qp, Gp) &= (x, p, q, \gamma)_+, \\ xm &= x_- = \phi / m, \\ pm &= p_- = kp / m, \\ (xt, pt, Gt, at) &= (x, p, \gamma, a)_\perp, \\ pp &= p_+, \end{aligned}$$

We introduce the following notations for $\gamma^5 (F^* x)_\beta$

$$\begin{aligned}GFDp &= (\gamma^\beta \gamma^5 F^*_{\beta\mu})_+, \\ GFDt &= (\gamma^\beta \gamma^5 F^*_{\beta\mu})_\perp, \\ \text{Note that } (\gamma^\beta \gamma^5 F^*_{\beta\mu})_- &= 0. \end{aligned}$$

We also will use the conservation law

$$\begin{aligned}qm &= pm, \\ qt &= pt; \end{aligned}$$

Scalar products will take the form

$$\begin{aligned}x^2 &= 2 x_+ x_- - x_\perp^2 = 2 xp xm - xt^2, \\ (px) &= p_+ x_- + p_- x_+ - p_\perp x_\perp = pp xm + pm xp - pt xt, \\ (ax) &= -a_\perp x_\perp = -at pt, \\ \gamma p &= \gamma_- p_+ + \gamma_+ p_- - \gamma_\perp p_\perp = Gm \frac{s}{x_-} (p^2 + p_\perp^2) + Gp \frac{x_-}{2s} - Gt pt, \end{aligned}$$

where we used that $p_+ =$

$$(p^2 + p_\perp^2) / 2 p_- \text{ and the definition of the proper time } s = x_- / 2 p_-;$$

$$\begin{aligned}\gamma k &= \gamma_- k_+ = m Gm, \\ \gamma^\beta \gamma^5 (F^* x)_\beta &= GFDp xm - GFDt xt. \end{aligned}$$

The integration measure

$$\begin{aligned}dx_- &= d\phi / m \\ dX_- &= d\bar{\phi} / m \\ \int dX_- d^D x \dots &= (m)^{-2} \int d\bar{\phi} d^{D-2} x_\perp d\phi dx_+ \dots \end{aligned}$$

$$\text{remaining integrals : } \int d\bar{\phi} d^{D-2} x_\perp d\phi dx_+$$

In[40]:= **Coeff5 = Coeff4 / (m) ^ 2**

Phase5 = Collect[Phase4 /.

{xv2 → 2 xm xp - xt ^ 2,

ax → -at * xt,

Pair[Momentum[p, D], Momentum[x, D]] → pp xm + pm xp - pt * xt,

Pair[Momentum[q, D], Momentum[x, D]] → qp xm + qm xp - qt * xt,

Pair[Momentum[a, D], Momentum[p, D]] → -at pt} /.

{qm → pm, qt → pt} /. {xm → ϕ / m, pm → kp / m}, {xp, ϕ, xt}]

Matrix5 =

Collect[Expand[Matrix4 /. {DiracGamma[Momentum[x, D], D] → Gp xm + Gm * xp - Gt * xt,

DiracGamma[LorentzIndex[β, D], D].DiracGamma[5] ×

Pair[LorentzIndex[β, D], Momentum[FDx, D]] → GFDp xm - GFDt * xt,

Pair[Momentum[a, D], Momentum[x, D]] → -at * xt} /.

{xm → ϕ / m}], {J0, J1, J2, xt, xp}]

$$\text{Out[40]} = -\frac{2^{-D-2} e^{-\frac{1}{2} i \pi D} \phi^2 s^{-D/2} t^{-D/2}}{\pi^2 m}$$

$$\begin{aligned} \text{Out[41]} = & \phi \left(\frac{at e pt \Phi}{k \cdot p} + \frac{m^2 \xi^2 \Phi^2}{2 (k \cdot p)} + \frac{pp}{2 m} + \frac{qp}{2 m} \right) + xt (-at e \Phi - pt) + \frac{m^2 \xi^2 \phi^3}{24 (k \cdot p)} + \\ & xp \left(\frac{k \cdot p}{m} + \phi \left(-\frac{1}{2 m s} - \frac{1}{2 m t} \right) \right) - \frac{1}{12} m^2 \xi^2 s \phi^2 - m^2 s - \frac{pp \Phi}{m} + \frac{qp \Phi}{m} + xt^2 \left(\frac{1}{4 s} + \frac{1}{4 t} \right) \end{aligned}$$

$$\begin{aligned}
\text{Out[42]= } & \text{J1} \left(-\frac{m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} - \frac{m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} - \frac{i e \text{GFDp} \phi^2}{4 m^2 s (k \cdot p)} - \frac{1}{3} m s \xi^2 \gamma \cdot k \phi + \frac{\text{Gp} \phi}{2 m^2 s} + \frac{i e \sigma \text{F} \phi}{4 (k \cdot p)} - \frac{e \Phi \gamma \cdot a \phi}{2 m s (k \cdot p)} - \frac{i e \text{GFDp} \phi}{2 m^2} + \right. \\
& \left. \frac{m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} + \frac{\text{Gm xp}}{2 m s} + \frac{1}{2} i e s \sigma \text{F} + \text{xt} \left(\frac{i e \text{GFDt}}{2 m} + \frac{i e \phi \text{GFDt}}{4 m s (k \cdot p)} - \frac{\text{Gt}}{2 m s} - \frac{\text{at} e \Phi \gamma \cdot k}{2 m s (k \cdot p)} \right) - 1 - \frac{2 i t \gamma \cdot k}{m s \phi} \right) + \\
& \text{J0} \left(\frac{D m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} - \frac{m \xi^2 \gamma \cdot k \phi^3}{8 s (k \cdot p)^2} - \frac{D m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} + \frac{3 m \xi^2 \gamma \cdot k \phi^2}{2 (k \cdot p)} + \frac{i D e \text{GFDp} \phi^2}{4 m^2 s (k \cdot p)} - \frac{i e \text{GFDp} \phi^2}{2 m^2 s (k \cdot p)} + \right. \\
& \frac{1}{3} D m s \xi^2 \gamma \cdot k \phi - \frac{2}{3} m s \xi^2 \gamma \cdot k \phi - \frac{D \text{Gp} \phi}{2 m^2 s} + \frac{\text{Gp} \phi}{m^2 s} - \frac{i D e \sigma \text{F} \phi}{4 (k \cdot p)} + \frac{D e \Phi \gamma \cdot a \phi}{2 m s (k \cdot p)} - \frac{e \Phi \gamma \cdot a \phi}{m s (k \cdot p)} - \frac{i D e \text{GFDp} \phi}{2 m^2} + \\
& \frac{3 i e \text{GFDp} \phi}{m^2} - \frac{D m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} + \frac{m \xi^2 \Phi^2 \gamma \cdot k \phi}{2 s (k \cdot p)^2} + D + \left(\frac{\text{Gm}}{m s} - \frac{D \text{Gm}}{2 m s} \right) \text{xp} + \frac{1}{2} i D e s \sigma \text{F} - 2 i e s \sigma \text{F} + \\
& \left. \text{xt} \left(\frac{i D e \text{GFDt}}{2 m} - \frac{3 i e \text{GFDt}}{m} - \frac{i D e \phi \text{GFDt}}{4 m s (k \cdot p)} + \frac{i e \phi \text{GFDt}}{2 m s (k \cdot p)} + \frac{D \text{Gt}}{2 m s} - \frac{\text{Gt}}{m s} - \frac{\text{at} e \Phi \gamma \cdot k}{m s (k \cdot p)} + \frac{\text{at} D e \Phi \gamma \cdot k}{2 m s (k \cdot p)} \right) \right) + \\
& \text{J2} \left(-\frac{D m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} + \frac{3 m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} + \frac{D m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} - \frac{3 m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} - \frac{i D e \text{GFDp} \phi^2}{4 m^2 s (k \cdot p)} + \right. \\
& \frac{3 i e \text{GFDp} \phi^2}{4 m^2 s (k \cdot p)} - \frac{1}{3} D m s \xi^2 \gamma \cdot k \phi + m s \xi^2 \gamma \cdot k \phi + \frac{D \text{Gp} \phi}{2 m^2 s} - \frac{3 \text{Gp} \phi}{2 m^2 s} + \frac{i D e \sigma \text{F} \phi}{4 (k \cdot p)} - \\
& \frac{3 i e \sigma \text{F} \phi}{4 (k \cdot p)} - \frac{D e \Phi \gamma \cdot a \phi}{2 m s (k \cdot p)} + \frac{3 e \Phi \gamma \cdot a \phi}{2 m s (k \cdot p)} + \frac{i D e \text{GFDp} \phi}{2 m^2} - \frac{3 i e \text{GFDp} \phi}{2 m^2} + \\
& \frac{D m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} - \frac{3 m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} - D + \left(\frac{D \text{Gm}}{2 m s} - \frac{3 \text{Gm}}{2 m s} \right) \text{xp} - \frac{1}{2} i D e s \sigma \text{F} + \frac{3}{2} i e s \sigma \text{F} + \\
& \left. \text{xt} \left(-\frac{i D e \text{GFDt}}{2 m} + \frac{3 i e \text{GFDt}}{2 m} + \frac{i D e \phi \text{GFDt}}{4 m s (k \cdot p)} - \frac{3 i e \phi \text{GFDt}}{4 m s (k \cdot p)} - \frac{D \text{Gt}}{2 m s} + \frac{3 \text{Gt}}{2 m s} + \frac{3 \text{at} e \Phi \gamma \cdot k}{2 m s (k \cdot p)} - \frac{\text{at} D e \Phi \gamma \cdot k}{2 m s (k \cdot p)} \right) + \right. \\
& \left. 3 - \frac{2 i t \gamma \cdot k}{m s \phi} \right)
\end{aligned}$$

Integration over **$\int dx_+$ and then $\int d\phi$...**

The phase is linear in x_+ ,

the terms in the preexponent either do not depend or linear in x_+ ,

therefore we can perform the integration using the equality

$$\int dx_+ \left(\frac{1}{x_+} \right) \text{Exp} [i x_+ P(\phi)] = 2\pi \begin{pmatrix} \delta(P(\phi)) \\ -i \delta'(P(\phi)) \end{pmatrix}$$

To use it, we separate the terms linear in $x_p = x_+$.

Recall that J_k depends in ϕ , as

$$J_k = J_k[t, \chi_1 = \frac{\xi \phi}{2 m^2 t}],$$

In the next steps we will use the shorthand notation

$$J_k = J_k[\phi].$$

```
In[43]:= Matrix52 = Collect[
  Coefficient[Expand[Matrix5], Gm xp] Gm xp /.
    {J0 -> J0[phi], J1 -> J1[phi], J2 -> J2[phi]},
  {Gm xp, J0[phi], J1[phi], J2[phi]}]
Matrix51 = Collect[
  (Expand[Matrix5] /. {J0 -> J0[phi], J1 -> J1[phi], J2 -> J2[phi]}) - Expand[Matrix52],
  {J0[phi], J1[phi], J2[phi], xt, xp}]

Out[43]= Gm xp \left( J0(\phi) \left( \frac{1}{ms} - \frac{D}{2ms} \right) + J2(\phi) \left( \frac{D}{2ms} - \frac{3}{2ms} \right) + \frac{J1(\phi)}{2ms} \right)
```

$$\begin{aligned}
\text{Out[44]} = & \text{J1}(\phi) \left(-\frac{m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} - \frac{m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} - \frac{i e \text{GFDp} \phi^2}{4 m^2 s (k \cdot p)} - \frac{1}{3} m s \xi^2 \gamma \cdot k \phi + \frac{\text{Gp} \phi}{2 m^2 s} + \frac{i e \sigma \text{F} \phi}{4 (k \cdot p)} - \frac{e \Phi \gamma \cdot a \phi}{2 m s (k \cdot p)} - \frac{i e \text{GFDp} \phi}{2 m^2} + \right. \\
& \left. \frac{m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} + \frac{1}{2} i e s \sigma \text{F} + \text{xt} \left(\frac{i e \text{GFDt}}{2 m} + \frac{i e \phi \text{GFDt}}{4 m s (k \cdot p)} - \frac{\text{Gt}}{2 m s} - \frac{\text{at} e \Phi \gamma \cdot k}{2 m s (k \cdot p)} \right) - 1 - \frac{2 i t \gamma \cdot k}{m s \phi} \right) + \\
& \text{J0}(\phi) \left(\frac{D m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} - \frac{m \xi^2 \gamma \cdot k \phi^3}{8 s (k \cdot p)^2} - \frac{D m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} + \frac{3 m \xi^2 \gamma \cdot k \phi^2}{2 (k \cdot p)} + \frac{i D e \text{GFDp} \phi^2}{4 m^2 s (k \cdot p)} - \frac{i e \text{GFDp} \phi^2}{2 m^2 s (k \cdot p)} + \right. \\
& \frac{1}{3} D m s \xi^2 \gamma \cdot k \phi - \frac{2}{3} m s \xi^2 \gamma \cdot k \phi - \frac{D \text{Gp} \phi}{2 m^2 s} + \frac{\text{Gp} \phi}{m^2 s} - \frac{i D e \sigma \text{F} \phi}{4 (k \cdot p)} + \frac{D e \Phi \gamma \cdot a \phi}{2 m s (k \cdot p)} - \frac{e \Phi \gamma \cdot a \phi}{m s (k \cdot p)} - \\
& \frac{i D e \text{GFDp} \phi}{2 m^2} + \frac{3 i e \text{GFDp} \phi}{m^2} - \frac{D m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} + \frac{m \xi^2 \Phi^2 \gamma \cdot k \phi}{2 s (k \cdot p)^2} + D + \frac{1}{2} i D e s \sigma \text{F} - 2 i e s \sigma \text{F} + \\
& \left. \text{xt} \left(\frac{i D e \text{GFDt}}{2 m} - \frac{3 i e \text{GFDt}}{m} - \frac{i D e \phi \text{GFDt}}{4 m s (k \cdot p)} + \frac{i e \phi \text{GFDt}}{2 m s (k \cdot p)} + \frac{D \text{Gt}}{2 m s} - \frac{\text{Gt}}{m s} - \frac{\text{at} e \Phi \gamma \cdot k}{m s (k \cdot p)} + \frac{\text{at} D e \Phi \gamma \cdot k}{2 m s (k \cdot p)} \right) \right) + \\
& \text{J2}(\phi) \left(-\frac{D m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} + \frac{3 m \xi^2 \gamma \cdot k \phi^3}{16 s (k \cdot p)^2} + \frac{D m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} - \frac{3 m \xi^2 \gamma \cdot k \phi^2}{4 (k \cdot p)} - \frac{i D e \text{GFDp} \phi^2}{4 m^2 s (k \cdot p)} + \frac{3 i e \text{GFDp} \phi^2}{4 m^2 s (k \cdot p)} - \right. \\
& \frac{1}{3} D m s \xi^2 \gamma \cdot k \phi + m s \xi^2 \gamma \cdot k \phi + \frac{D \text{Gp} \phi}{2 m^2 s} - \frac{3 \text{Gp} \phi}{2 m^2 s} + \frac{i D e \sigma \text{F} \phi}{4 (k \cdot p)} - \frac{3 i e \sigma \text{F} \phi}{4 (k \cdot p)} - \frac{D e \Phi \gamma \cdot a \phi}{2 m s (k \cdot p)} + \frac{3 e \Phi \gamma \cdot a \phi}{2 m s (k \cdot p)} + \\
& \frac{i D e \text{GFDp} \phi}{2 m^2} - \frac{3 i e \text{GFDp} \phi}{2 m^2} + \frac{D m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} - \frac{3 m \xi^2 \Phi^2 \gamma \cdot k \phi}{4 s (k \cdot p)^2} - D - \frac{1}{2} i D e s \sigma \text{F} + \frac{3}{2} i e s \sigma \text{F} + \\
& \left. \text{xt} \left(-\frac{i D e \text{GFDt}}{2 m} + \frac{3 i e \text{GFDt}}{2 m} + \frac{i D e \phi \text{GFDt}}{4 m s (k \cdot p)} - \frac{3 i e \phi \text{GFDt}}{4 m s (k \cdot p)} - \frac{D \text{Gt}}{2 m s} + \frac{3 \text{Gt}}{2 m s} + \frac{3 \text{at} e \Phi \gamma \cdot k}{2 m s (k \cdot p)} - \frac{\text{at} D e \Phi \gamma \cdot k}{2 m s (k \cdot p)} \right) + \right. \\
& \left. 3 - \frac{2 i t \gamma \cdot k}{m s \phi} \right)
\end{aligned}$$

In[45]:= **Phase5xp =**

Collect[Simplify[Coefficient[Phase5, xp] /. {t → s ω / (s - ω)}], {kp}, Simplify]

φ0 = Collect[φ /. Solve[Phase5xp == 0, φ][[1]], {kp}, Simplify]

dP = Simplify[D[Phase5xp, φ]] /. {φ → φ0}

AbsdP = -dP

Phase5noxp = Collect[Phase5 /. {xp → 0} /. {t → s ω / (s - ω)}, {xp, φ, xt}, Simplify]

$$\text{Out[45]} = \frac{k \cdot p}{m} - \frac{\phi}{2 m \omega}$$

$$\text{Out[46]} = 2 \omega (k \cdot p)$$

$$\text{Out[47]} = -\frac{1}{2 m \omega}$$

$$\text{Out[48]} = \frac{1}{2 m \omega}$$

$$\text{Out[49]} = \frac{1}{2} \phi \left(\frac{\Phi (2 \text{at} e \text{pt} + m^2 \xi^2 \Phi)}{k \cdot p} + \frac{\text{pp} + \text{qp}}{m} \right) + \text{xt} (-\text{at} e \Phi - \text{pt}) + \frac{m^2 \xi^2 \phi^3}{24 (k \cdot p)} - \frac{m^3 s + \text{pp} \Phi - \text{qp} \Phi}{m} - \frac{1}{12} m^2 \xi^2 s \phi^2 + \frac{\text{xt}^2}{4 \omega}$$

Now we can perform the integrations

$$\int dx_+ \left(\frac{1}{x_+} \right) \text{Exp} [i x_+ P(\phi)] =$$

$$2\pi \left(\begin{array}{c} \delta(P(\phi)) \\ -i\delta'(P(\phi)) \end{array} \right) = 2\pi \left(\begin{array}{c} \delta(\phi - \phi_0) \frac{1}{|P'(\phi_0)|} \\ -i\delta(\phi - \phi_0) \frac{1}{|P'(\phi_0)|} \left(-\frac{d}{d\phi} \frac{1}{P'(\phi)} \cdot \right) \end{array} \right)$$

Here $\left(-\frac{d}{d\phi} \frac{1}{P'(\phi)} \cdot \right)$ is an operator,
acting on a function in the place of ".". Then,

$$\int d\phi 2\pi \left(\begin{array}{c} \delta(\phi - \phi_0) \frac{1}{|P'(\phi_0)|} \\ -i\delta(\phi - \phi_0) \frac{1}{|P'(\phi_0)|} \left(-\frac{d}{d\phi} \frac{1}{P'(\phi)} \cdot \right) \end{array} \right) f(\phi) e^{ig(\phi)} =$$

$$\frac{2\pi}{|P'(\phi_0)|} e^{ig(\phi_0)} \left(\begin{array}{c} f(\phi_0) \\ i\frac{d}{d\phi} \left(f(\phi) \frac{1}{P'(\phi)} \right) |_{\phi_0} - f(\phi_0) \frac{g'(\phi_0)}{P'(\phi_0)} \end{array} \right)$$

$$P(\phi) = \frac{(k \cdot p)}{m} - \frac{\phi}{2m\omega},$$

$$P'(\phi) = -\frac{1}{2m\omega},$$

$$\omega^{-1} = s^{-1} + t^{-1}, \quad \omega = st / (s + t)$$

Note that we will take the derivative of $J_k[\phi]$ in ϕ too;

We introduce the shorthand

$$DJ_k \equiv \frac{d}{d\phi} J_k[\phi] \equiv \frac{d}{d\phi} J_k(t, \chi_1(\phi))$$

The remaining integrals : $\int d\Phi d^{D-2}x_\perp$

```
In[50]:= Coeff6 = Coeff5 * 2 * Pi / AbsdP
```

```
Phase6 = Collect[
```

```
  Expand[Simplify[Phase5 /. {xp -> 0} /. {phi -> phi0} /. {t -> s*omega/(s-omega)}]], {xt, Phi}]
```

```
Out[50]= -\frac{2^{-D} e^{-\frac{1}{2} i \pi D} e^2 \omega s^{-D/2} t^{-D/2}}{\pi}
```

```
Out[51]= \frac{1}{3} m^2 \xi^2 \omega^3 (k \cdot p)^2 - \frac{1}{3} m^2 \xi^2 s \omega^2 (k \cdot p)^2 + \Phi \left( 2 \text{at} e p t \omega - \frac{p p}{m} + \frac{q p}{m} \right) +
```

$$x t (-\text{at} e \Phi - p t) + \frac{p p \omega (k \cdot p)}{m} + \frac{q p \omega (k \cdot p)}{m} + m^2 \xi^2 \Phi^2 \omega - m^2 s + \frac{x t^2}{4 \omega}$$

Note that Matrix52 produces terms that are proportional to J_k and $\frac{dJ_k}{s\phi}$.

We combine the ones that are proportional to J_k together in Matrix61, and leave the rest in Matrix62

We also again denote

$$J_k = J_k\left[t, \frac{\xi \phi_0}{2 m^2 t}\right]$$

```
In[52]:= Matrix61 = Collect[
  Expand[Matrix51 - (Matrix52 /. {xp → D[Phase5noxp, ϕ] / dP})] /. {J0[ϕ] → J0,
    J1[ϕ] → J1, J2[ϕ] → J2} /. {ϕ → ϕ0}, {J0, J1, J2, xt, Gm, Gp, GFDp, pp}]
Matrix62 = ID[1 / dP * Matrix52, ϕ] /. {xp → 1} /. {ϕ → ϕ0}
```

$$\begin{aligned}
\text{Out[52]} = & \text{J1} \left(-\frac{m \xi^2 \gamma \cdot k (k \cdot p) \omega^3}{2 s} - m \xi^2 \gamma \cdot k (k \cdot p) \omega^2 + \frac{1}{2} i e \sigma F \omega - \right. \\
& \frac{e \Phi \gamma \cdot a \omega}{m s} - \frac{2}{3} m s \xi^2 \gamma \cdot k (k \cdot p) \omega + \frac{\text{Gp}(k \cdot p) \omega}{m^2 s} + \frac{m \xi^2 \Phi^2 \gamma \cdot k \omega}{2 s (k \cdot p)} + \frac{1}{2} i e s \sigma F + \\
& \text{xt} \left(\frac{i e \omega \text{GFDt}}{2 m s} + \frac{i e \text{GFDt}}{2 m} - \frac{\text{Gt}}{2 m s} - \frac{\text{at} e \Phi \gamma \cdot k}{2 m s (k \cdot p)} \right) + \text{GFDp} \left(-\frac{i e (k \cdot p) \omega^2}{m^2 s} - \frac{i e (k \cdot p) \omega}{m^2} \right) + \\
& \text{Gm} \left(\frac{m^2 \xi^2 (k \cdot p) \omega^3}{2 s} - \frac{1}{3} m^2 \xi^2 (k \cdot p) \omega^2 + \frac{\text{pp} \omega}{2 m s} + \frac{\text{qp} \omega}{2 m s} + \frac{m^2 \xi^2 \Phi^2 \omega}{2 s (k \cdot p)} + \frac{\text{at} e \text{pt} \Phi \omega}{s (k \cdot p)} \right) - 1 - \frac{i t \gamma \cdot k}{m s (k \cdot p) \omega} \Big) + \\
& \text{J0} \left(\frac{D m \xi^2 \gamma \cdot k (k \cdot p) \omega^3}{2 s} - \frac{m \xi^2 \gamma \cdot k (k \cdot p) \omega^3}{s} - D m \xi^2 \gamma \cdot k (k \cdot p) \omega^2 + 6 m \xi^2 \gamma \cdot k (k \cdot p) \omega^2 - \right. \\
& \frac{1}{2} i D e \sigma F \omega + \frac{D e \Phi \gamma \cdot a \omega}{m s} - \frac{2 e \Phi \gamma \cdot a \omega}{m s} + \frac{2}{3} D m s \xi^2 \gamma \cdot k (k \cdot p) \omega - \\
& \frac{4}{3} m s \xi^2 \gamma \cdot k (k \cdot p) \omega - \frac{D m \xi^2 \Phi^2 \gamma \cdot k \omega}{2 s (k \cdot p)} + \frac{m \xi^2 \Phi^2 \gamma \cdot k \omega}{s (k \cdot p)} + D + \frac{1}{2} i D e s \sigma F - 2 i e s \sigma F + \\
& \text{xt} \left(-\frac{i D e \omega \text{GFDt}}{2 m s} + \frac{i e \omega \text{GFDt}}{m s} + \frac{i D e \text{GFDt}}{2 m} - \frac{3 i e \text{GFDt}}{m} + \frac{D \text{Gt}}{2 m s} - \frac{\text{Gt}}{m s} - \frac{\text{at} e \Phi \gamma \cdot k}{m s (k \cdot p)} + \frac{\text{at} D e \Phi \gamma \cdot k}{2 m s (k \cdot p)} \right) + \\
& \text{Gp} \left(\frac{2 \omega (k \cdot p)}{m^2 s} - \frac{D \omega (k \cdot p)}{m^2 s} \right) + \text{GFDp} \left(\frac{i D e (k \cdot p) \omega^2}{m^2 s} - \frac{2 i e (k \cdot p) \omega^2}{m^2 s} - \frac{i D e (k \cdot p) \omega}{m^2} + \frac{6 i e (k \cdot p) \omega}{m^2} \right) + \\
& \text{Gm} \left(-\frac{D m^2 \xi^2 (k \cdot p) \omega^3}{2 s} + \frac{m^2 \xi^2 (k \cdot p) \omega^3}{s} + \frac{1}{3} D m^2 \xi^2 (k \cdot p) \omega^2 - \frac{2}{3} m^2 \xi^2 (k \cdot p) \omega^2 - \frac{D \text{qp} \omega}{2 m s} + \right. \\
& \left. \frac{\text{qp} \omega}{m s} - \frac{D m^2 \xi^2 \Phi^2 \omega}{2 s (k \cdot p)} + \frac{m^2 \xi^2 \Phi^2 \omega}{s (k \cdot p)} + \frac{2 \text{at} e \text{pt} \Phi \omega}{s (k \cdot p)} - \frac{\text{at} D e \text{pt} \Phi \omega}{s (k \cdot p)} + \text{pp} \left(\frac{\omega}{m s} - \frac{D \omega}{2 m s} \right) \right) \Big) + \\
& \text{J2} \left(-\frac{D m \xi^2 \gamma \cdot k (k \cdot p) \omega^3}{2 s} + \frac{3 m \xi^2 \gamma \cdot k (k \cdot p) \omega^3}{2 s} + D m \xi^2 \gamma \cdot k (k \cdot p) \omega^2 - 3 m \xi^2 \gamma \cdot k (k \cdot p) \omega^2 + \right. \\
& \frac{1}{2} i D e \sigma F \omega - \frac{3}{2} i e \sigma F \omega - \frac{D e \Phi \gamma \cdot a \omega}{m s} + \frac{3 e \Phi \gamma \cdot a \omega}{m s} - \frac{2}{3} D m s \xi^2 \gamma \cdot k (k \cdot p) \omega + \\
& 2 m s \xi^2 \gamma \cdot k (k \cdot p) \omega + \frac{D m \xi^2 \Phi^2 \gamma \cdot k \omega}{2 s (k \cdot p)} - \frac{3 m \xi^2 \Phi^2 \gamma \cdot k \omega}{2 s (k \cdot p)} - D - \frac{1}{2} i D e s \sigma F + \frac{3}{2} i e s \sigma F + \\
& \text{xt} \left(\frac{i D e \omega \text{GFDt}}{2 m s} - \frac{3 i e \omega \text{GFDt}}{2 m s} - \frac{i D e \text{GFDt}}{2 m} + \frac{3 i e \text{GFDt}}{2 m} - \frac{D \text{Gt}}{2 m s} + \frac{3 \text{Gt}}{2 m s} + \frac{3 \text{at} e \Phi \gamma \cdot k}{2 m s (k \cdot p)} - \frac{\text{at} D e \Phi \gamma \cdot k}{2 m s (k \cdot p)} \right) + \\
& \text{Gp} \left(\frac{D \omega (k \cdot p)}{m^2 s} - \frac{3 \omega (k \cdot p)}{m^2 s} \right) + \text{GFDp} \left(-\frac{i D e (k \cdot p) \omega^2}{m^2 s} + \frac{3 i e (k \cdot p) \omega^2}{m^2 s} + \frac{i D e (k \cdot p) \omega}{m^2} - \frac{3 i e (k \cdot p) \omega}{m^2} \right) + \\
& \text{Gm} \left(\frac{D m^2 \xi^2 (k \cdot p) \omega^3}{2 s} - \frac{3 m^2 \xi^2 (k \cdot p) \omega^3}{2 s} - \frac{1}{3} D m^2 \xi^2 (k \cdot p) \omega^2 + m^2 \xi^2 (k \cdot p) \omega^2 + \frac{D \text{qp} \omega}{2 m s} - \frac{3 \text{qp} \omega}{2 m s} + \right. \\
& \left. \frac{D m^2 \xi^2 \Phi^2 \omega}{2 s (k \cdot p)} - \frac{3 m^2 \xi^2 \Phi^2 \omega}{2 s (k \cdot p)} - \frac{3 \text{at} e \text{pt} \Phi \omega}{s (k \cdot p)} + \frac{\text{at} D e \text{pt} \Phi \omega}{s (k \cdot p)} + \text{pp} \left(\frac{D \omega}{2 m s} - \frac{3 \omega}{2 m s} \right) \right) + 3 - \frac{i t \gamma \cdot k}{m s (k \cdot p) \omega} \Big) \\
& \text{Out[53]} = -2 i \text{Gm} m \omega \left(\left(\frac{1}{m s} - \frac{D}{2 m s} \right) \text{J0}'(2 \omega (k \cdot p)) + \left(\frac{D}{2 m s} - \frac{3}{2 m s} \right) \text{J2}'(2 \omega (k \cdot p)) + \frac{\text{J1}'(2 \omega (k \cdot p))}{2 m s} \right)
\end{aligned}$$

Let us rewrite Matrix62

After the last integration

$$J_k = J_k \left(t, \frac{\xi \phi_0}{2 m^2 t} \right);$$

$$J_k'(\phi_0) = \frac{\partial J_k \left(t, \frac{\xi \phi}{2 m^2 t} \right)}{\partial \phi} \Big|_{\phi_0} = \frac{\partial}{\partial \chi_1} J_k \left(t, \chi_1(\phi_0) \right) * \frac{\xi}{2 m^2 t};$$

$$\chi_1(\phi_0) = \frac{\xi \phi_0}{2 m^2 t} = \frac{\xi \omega(k \cdot p)}{m^2 t};$$

In this step,

we use the initial assumption that J_0 does not depend on χ_1 , therefore

$$\frac{\partial}{\partial \chi_1} J_0 = 0.$$

We denote

$$dJ_k d\chi_l = \frac{\partial}{\partial \chi_l} J_k \left(t, \chi_l(\phi_0) \right), \quad k = 1, 2$$

```
In[54]:= Coeff7 = Coeff6;
Phase7 = Phase6;
Matrix71 = Matrix61;

In[57]:=  $\chi_l \phi = \xi \phi / 2 / m^2 / t$ 
 $\chi_l \phi_0 = \chi_l \phi /. \{\phi \rightarrow \phi_0\}$ 
 $d\chi_l d\phi = D[\chi_l \phi, \phi] /. \{\phi \rightarrow \phi_0\}$ 
Matrix72d =
Collect[Matrix62 /. {J0'[2 ω kp] → dJ0dχl * dχldφ, J1'[2 ω kp] → dJ1dχl * dχldφ,
J2'[2 ω kp] → dJ2dχl * dχldφ} /. {dJ0dχl → 0}, {dJ0dχl, dJ1dχl, dJ2dχl}]

Out[57]=  $\frac{\xi \phi}{2 m^2 t}$ 

Out[58]=  $\frac{\xi \omega(k \cdot p)}{m^2 t}$ 

Out[59]=  $\frac{\xi}{2 m^2 t}$ 

Out[60]=  $-\frac{i dJ2d\chi_1 G m \xi \omega \left( \frac{D}{2 m s} - \frac{3}{2 m s} \right)}{m t} - \frac{i dJ1d\chi_1 G m \xi \omega}{2 m^2 s t}$ 
```

In[61]:= **Matrix7 = Collect[Matrix71 + Matrix72d,**

{J0, J1, J2, dJ0dχl, dJ1dχl, dJ2dχl, xt, Gm, Gp, GFDp, pp}]

$$\begin{aligned}
 \text{Out[61]} = & -\frac{i dJ2d\chi l Gm \left(\frac{D}{2ms} - \frac{3}{2ms} \right) \xi \omega}{m t} - \frac{i dJ1d\chi l Gm \xi \omega}{2 m^2 s t} + \\
 & J1 \left(-\frac{m \xi^2 \gamma \cdot k (k \cdot p) \omega^3}{2 s} - m \xi^2 \gamma \cdot k (k \cdot p) \omega^2 + \frac{1}{2} i e \sigma F \omega - \frac{e \Phi \gamma \cdot a \omega}{m s} - \right. \\
 & \quad \frac{2}{3} m s \xi^2 \gamma \cdot k (k \cdot p) \omega + \frac{Gp (k \cdot p) \omega}{m^2 s} + \frac{m \xi^2 \Phi^2 \gamma \cdot k \omega}{2 s (k \cdot p)} + \frac{1}{2} i e s \sigma F + \\
 & \quad xt \left(\frac{i e \omega GFDt}{2 m s} + \frac{i e GFDt}{2 m} - \frac{Gt}{2 m s} - \frac{at e \Phi \gamma \cdot k}{2 m s (k \cdot p)} \right) + GFDp \left(-\frac{i e (k \cdot p) \omega^2}{m^2 s} - \frac{i e (k \cdot p) \omega}{m^2} \right) + \\
 & \quad Gm \left(\frac{m^2 \xi^2 (k \cdot p) \omega^3}{2 s} - \frac{1}{3} m^2 \xi^2 (k \cdot p) \omega^2 + \frac{pp \omega}{2 m s} + \frac{qp \omega}{2 m s} + \frac{m^2 \xi^2 \Phi^2 \omega}{2 s (k \cdot p)} + \frac{at e pt \Phi \omega}{s (k \cdot p)} \right) - 1 - \frac{i t \gamma \cdot k}{m s (k \cdot p) \omega} \Bigg) + \\
 & J0 \left(\frac{D m \xi^2 \gamma \cdot k (k \cdot p) \omega^3}{2 s} - \frac{m \xi^2 \gamma \cdot k (k \cdot p) \omega^3}{s} - D m \xi^2 \gamma \cdot k (k \cdot p) \omega^2 + 6 m \xi^2 \gamma \cdot k (k \cdot p) \omega^2 - \right. \\
 & \quad \frac{1}{2} i D e \sigma F \omega + \frac{D e \Phi \gamma \cdot a \omega}{m s} - \frac{2 e \Phi \gamma \cdot a \omega}{m s} + \frac{2}{3} D m s \xi^2 \gamma \cdot k (k \cdot p) \omega - \\
 & \quad \frac{4}{3} m s \xi^2 \gamma \cdot k (k \cdot p) \omega - \frac{D m \xi^2 \Phi^2 \gamma \cdot k \omega}{2 s (k \cdot p)} + \frac{m \xi^2 \Phi^2 \gamma \cdot k \omega}{s (k \cdot p)} + D + \frac{1}{2} i D e s \sigma F - 2 i e s \sigma F + \\
 & \quad xt \left(-\frac{i D e \omega GFDt}{2 m s} + \frac{i e \omega GFDt}{m s} + \frac{i D e GFDt}{2 m} - \frac{3 i e GFDt}{m} + \frac{D Gt}{2 m s} - \frac{Gt}{m s} - \frac{at e \Phi \gamma \cdot k}{m s (k \cdot p)} + \frac{at D e \Phi \gamma \cdot k}{2 m s (k \cdot p)} \right) + \\
 & \quad Gp \left(\frac{2 \omega (k \cdot p)}{m^2 s} - \frac{D \omega (k \cdot p)}{m^2 s} \right) + GFDp \left(\frac{i D e (k \cdot p) \omega^2}{m^2 s} - \frac{2 i e (k \cdot p) \omega^2}{m^2 s} - \frac{i D e (k \cdot p) \omega}{m^2} + \frac{6 i e (k \cdot p) \omega}{m^2} \right) + \\
 & \quad Gm \left(-\frac{D m^2 \xi^2 (k \cdot p) \omega^3}{2 s} + \frac{m^2 \xi^2 (k \cdot p) \omega^3}{s} + \frac{1}{3} D m^2 \xi^2 (k \cdot p) \omega^2 - \frac{2}{3} m^2 \xi^2 (k \cdot p) \omega^2 - \frac{D qp \omega}{2 m s} + \right. \\
 & \quad \left. \frac{qp \omega}{m s} - \frac{D m^2 \xi^2 \Phi^2 \omega}{2 s (k \cdot p)} + \frac{m^2 \xi^2 \Phi^2 \omega}{s (k \cdot p)} + \frac{2 at e pt \Phi \omega}{s (k \cdot p)} - \frac{at D e pt \Phi \omega}{s (k \cdot p)} + pp \left(\frac{\omega}{m s} - \frac{D \omega}{2 m s} \right) \right) \Bigg) + \\
 & J2 \left(-\frac{D m \xi^2 \gamma \cdot k (k \cdot p) \omega^3}{2 s} + \frac{3 m \xi^2 \gamma \cdot k (k \cdot p) \omega^3}{2 s} + D m \xi^2 \gamma \cdot k (k \cdot p) \omega^2 - 3 m \xi^2 \gamma \cdot k (k \cdot p) \omega^2 + \right. \\
 & \quad \frac{1}{2} i D e \sigma F \omega - \frac{3}{2} i e \sigma F \omega - \frac{D e \Phi \gamma \cdot a \omega}{m s} + \frac{3 e \Phi \gamma \cdot a \omega}{m s} - \frac{2}{3} D m s \xi^2 \gamma \cdot k (k \cdot p) \omega + \\
 & \quad 2 m s \xi^2 \gamma \cdot k (k \cdot p) \omega + \frac{D m \xi^2 \Phi^2 \gamma \cdot k \omega}{2 s (k \cdot p)} - \frac{3 m \xi^2 \Phi^2 \gamma \cdot k \omega}{2 s (k \cdot p)} - D - \frac{1}{2} i D e s \sigma F + \frac{3}{2} i e s \sigma F + \\
 & \quad xt \left(\frac{i D e \omega GFDt}{2 m s} - \frac{3 i e \omega GFDt}{2 m s} - \frac{i D e GFDt}{2 m} + \frac{3 i e GFDt}{2 m} - \frac{D Gt}{2 m s} + \frac{3 Gt}{2 m s} + \frac{3 at e \Phi \gamma \cdot k}{2 m s (k \cdot p)} - \frac{at D e \Phi \gamma \cdot k}{2 m s (k \cdot p)} \right) + \\
 & \quad Gp \left(\frac{D \omega (k \cdot p)}{m^2 s} - \frac{3 \omega (k \cdot p)}{m^2 s} \right) + GFDp \left(-\frac{i D e (k \cdot p) \omega^2}{m^2 s} + \frac{3 i e (k \cdot p) \omega^2}{m^2 s} + \frac{i D e (k \cdot p) \omega}{m^2} - \frac{3 i e (k \cdot p) \omega}{m^2} \right) + \\
 & \quad Gm \left(\frac{D m^2 \xi^2 (k \cdot p) \omega^3}{2 s} - \frac{3 m^2 \xi^2 (k \cdot p) \omega^3}{2 s} - \frac{1}{3} D m^2 \xi^2 (k \cdot p) \omega^2 + m^2 \xi^2 (k \cdot p) \omega^2 + \frac{D qp \omega}{2 m s} - \frac{3 qp \omega}{2 m s} + \right. \\
 & \quad \left. \frac{D m^2 \xi^2 \Phi^2 \omega}{2 s (k \cdot p)} - \frac{3 m^2 \xi^2 \Phi^2 \omega}{2 s (k \cdot p)} - \frac{3 at e pt \Phi \omega}{s (k \cdot p)} + \frac{at D e pt \Phi \omega}{s (k \cdot p)} + pp \left(\frac{D \omega}{2 m s} - \frac{3 \omega}{2 m s} \right) \right) + 3 - \frac{i t \gamma \cdot k}{m s (k \cdot p) \omega} \Bigg)
 \end{aligned}$$

Integration over

$$\int d^{D-2} x_{\perp} \dots :$$

The integral is gaussian

$$\begin{aligned} I_0 &= \int d^{D-2} x_{\perp} \text{Exp} \left[I A x_{\perp}^2 + I (J \cdot x_{\perp}) \right] = \\ &= \text{Exp} \left[I \frac{\pi}{2} \frac{D-2}{2} \right] \pi^{\frac{D-2}{2}} (\det A)^{-\frac{1}{2}} \text{Exp} \left[-I \frac{1}{4} J \cdot A^{-1} \cdot J \right] \end{aligned}$$

The preexponent contains terms linear in $x_{\perp}$, so we will also need

$$\begin{aligned} I_1 &= \int d^{D-2} x_{\perp} x_{\perp} \text{Exp} \left[I A x_{\perp}^2 + I (J \cdot x_{\perp}) \right] = \\ &= -\frac{1}{2} (A^{-1} \cdot J) I_0; \end{aligned}$$

$$\begin{aligned} I_2 &= \int d^{D-2} x_{\perp} x_{\perp}^2 \text{Exp} \left[I A x_{\perp}^2 + I (J \cdot x_{\perp}) \right] = \\ &= \left[I \frac{1}{2} \text{Tr} A^{-1} + \left(-\frac{1}{2} (A^{-1} \cdot J) \right)^2 \right] I_0 \end{aligned}$$

The remaining integrals : $\int d\Omega$

```
In[62]:= Amatr = Coefficient[Phase7, xt^2]
DetA = Amatr ^ (D - 2)
Jvec = Coefficient[Phase7, xt]
Jvec2 = Collect[Expand[Jvec^2], {ϕ}]
CI0 = Exp[I Pi / 2 (D / 2 - 1)] Pi ^ (D / 2 - 1) (Amatr) ^ (- (D - 2) / 2)
PhaseI0 = -1 / 4 (1 / Amatr) Jvec2
xtQuadraticSubst = {xt^2 -> (I * 1 / 2 / Amatr * (D - 2) + (-1 / 2 Amatr ^ (-1) Jvec) ^ 2)}
xtLinearSubst = {xt -> -1 / 2 Amatr ^ (-1) Jvec}
```

$$\text{Out[62]= } \frac{1}{4 \omega}$$

$$\text{Out[63]= } 4^{2-D} \left(\frac{1}{\omega} \right)^{D-2}$$

$$\text{Out[64]= } -at e^{\Phi} - pt$$

$$\text{Out[65]= } at^2 e^2 \Phi^2 + 2 at e pt \Phi + pt^2$$

$$\text{Out[66]= } 2^{D-2} e^{\frac{1}{2} i \pi \left(\frac{D}{2} - 1 \right)} \pi^{\frac{D}{2} - 1} \left(\frac{1}{\omega} \right)^{\frac{2-D}{2}}$$

$$\text{Out[67]= } -(\omega (at^2 e^2 \Phi^2 + 2 at e pt \Phi + pt^2))$$

$$\text{Out}[68]= \{xt^2 \rightarrow 4 \omega^2 (-at e \Phi - pt)^2 + 2 i (D - 2) \omega\}$$

$$\text{Out}[69]= \{xt \rightarrow -2 \omega (-at e \Phi - pt)\}$$

We find that

$$A = \frac{1}{4 \omega},$$

$$J = -ea_{\perp} \Phi - p_{\perp},$$

$$\det A = \left(\frac{1}{4 \omega} \right)^{D-2},$$

$$A^{-1} = 4 \omega,$$

Also we use the following equalities for further simplifications :

$$a^{\mu} \gamma^{\beta} \gamma^5 F^*_{\beta\mu} = -a_{\perp} \left(\gamma^{\beta} \gamma^5 F^*_{\beta\mu} \right)_{\perp} = at \text{GFDt} = 0, \text{ as } a^{\mu} F^*_{\beta\mu} = 0,$$

$$at \text{Gt} = -(\gamma a),$$

$$at \text{pt} = (ap),$$

$$at^2 = -a^2 = \xi^2 m^2 / e^2$$

```
In[70]:= Coeff8 = Simplify[Coeff7 * CI0, Assumptions -> {\omega > 0, m > 0, t > 0, s > 0}]
Phase8 = Collect[(Phase7 /. {xt -> 0}) + PhaseI0, \Phi] /. {at^2 -> \xi^2 m^2 / e^2}
Matrix8 = Collect[
  Expand[
    Matrix7 /. xtQuadraticSubst /. xtLinearSubst
  ] /. {at GFDt -> 0, at Gt -> -DiracGamma[Momentum[a, D], D],
    at^2 -> \xi^2 m^2 / e^2, at pt -> -Pair[Momentum[a, D], Momentum[p, D]]},
  {J0, J1, J2, dJ0d\chi, dJ1d\chi, dJ2d\chi, \Phi, Gm, Gp, GFDp, pp}]
```

$$\text{Out}[70]= \frac{1}{4} i e^{-\frac{1}{4} i \pi D} \pi^{\frac{D}{2}-2} e^2 \left(\frac{st}{\omega} \right)^{-D/2}$$

$$\text{Out}[71]= \frac{1}{3} m^2 \xi^2 \omega^3 (k \cdot p)^2 - \frac{1}{3} m^2 \xi^2 s \omega^2 (k \cdot p)^2 + \frac{pp \omega (k \cdot p)}{m} + \frac{qp \omega (k \cdot p)}{m} - m^2 s + \Phi \left(\frac{qp}{m} - \frac{pp}{m} \right) - pt^2 \omega$$

$$\begin{aligned}
\text{Out}[72] = & -\frac{i \, dJ1 d\chi_1 \, Gm \, \xi \, \omega}{2 \, m^2 \, s \, t} + dJ2 d\chi_1 \, Gm \left(\frac{3 \, i \, \xi \, \omega}{2 \, m^2 \, s \, t} - \frac{i \, D \, \xi \, \omega}{2 \, m^2 \, s \, t} \right) + \\
& J1 \left(-\frac{m \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega^3}{2 \, s} - m \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega^2 + \frac{i \, e \, GFDt \, pt \, \omega^2}{m \, s} + \frac{i \, e \, GFDt \, pt \, \omega}{m} + \frac{1}{2} i \, e \, \sigma F \, \omega - \right. \\
& \frac{2}{3} m \, s \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega + \frac{Gp \, (k \cdot p) \, \omega}{m^2 \, s} - \frac{Gt \, pt \, \omega}{m \, s} + \frac{1}{2} i \, e \, s \, \sigma F + \Phi^2 \left(\frac{Gm \, m^2 \, \xi^2 \, \omega}{2 \, s \, (k \cdot p)} - \frac{m \, \xi^2 \, \omega \, \gamma \cdot k}{2 \, s \, (k \cdot p)} \right) + \\
& \Phi \left(\frac{e \, \omega \, \gamma \cdot k \, (a \cdot p)}{m \, s \, (k \cdot p)} - \frac{e \, Gm \, \omega \, (a \cdot p)}{s \, (k \cdot p)} \right) + GFDp \left(-\frac{i \, e \, (k \cdot p) \, \omega^2}{m^2 \, s} - \frac{i \, e \, (k \cdot p) \, \omega}{m^2} \right) + \\
& Gm \left(\frac{m^2 \, \xi^2 \, (k \cdot p) \, \omega^3}{2 \, s} - \frac{1}{3} m^2 \, \xi^2 \, (k \cdot p) \, \omega^2 + \frac{pp \, \omega}{2 \, m \, s} + \frac{qp \, \omega}{2 \, m \, s} \right) - 1 - \frac{i \, t \, \gamma \cdot k}{m \, s \, (k \cdot p) \, \omega} \Big) + \\
& J0 \left(\frac{D \, m \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega^3}{2 \, s} - \frac{m \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega^3}{s} - D \, m \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega^2 + 6 \, m \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega^2 - \right. \\
& \frac{i \, D \, e \, GFDt \, pt \, \omega^2}{m \, s} + \frac{2 \, i \, e \, GFDt \, pt \, \omega^2}{m \, s} + \frac{i \, D \, e \, GFDt \, pt \, \omega}{m} - \frac{6 \, i \, e \, GFDt \, pt \, \omega}{m} - \frac{1}{2} i \, D \, e \, \sigma F \, \omega + \\
& \frac{2}{3} D \, m \, s \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega - \frac{4}{3} m \, s \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega + \frac{D \, Gt \, pt \, \omega}{m \, s} - \frac{2 \, Gt \, pt \, \omega}{m \, s} + D + \\
& \frac{1}{2} i \, D \, e \, s \, \sigma F - 2 \, i \, e \, s \, \sigma F + \Phi^2 \left(\frac{D \, m \, \omega \, \gamma \cdot k \, \xi^2}{2 \, s \, (k \cdot p)} - \frac{m \, \omega \, \gamma \cdot k \, \xi^2}{s \, (k \cdot p)} + Gm \left(\frac{m^2 \, \xi^2 \, \omega}{s \, (k \cdot p)} - \frac{D \, m^2 \, \xi^2 \, \omega}{2 \, s \, (k \cdot p)} \right) \right) + \\
& \Phi \left(-\frac{D \, e \, \omega \, \gamma \cdot k \, (a \cdot p)}{m \, s \, (k \cdot p)} + \frac{2 \, e \, \omega \, \gamma \cdot k \, (a \cdot p)}{m \, s \, (k \cdot p)} + Gm \left(\frac{D \, e \, \omega \, (a \cdot p)}{s \, (k \cdot p)} - \frac{2 \, e \, \omega \, (a \cdot p)}{s \, (k \cdot p)} \right) \right) + Gp \left(\frac{2 \, \omega \, (k \cdot p)}{m^2 \, s} - \frac{D \, \omega \, (k \cdot p)}{m^2 \, s} \right) + \\
& GFDp \left(\frac{i \, D \, e \, (k \cdot p) \, \omega^2}{m^2 \, s} - \frac{2 \, i \, e \, (k \cdot p) \, \omega^2}{m^2 \, s} - \frac{i \, D \, e \, (k \cdot p) \, \omega}{m^2} + \frac{6 \, i \, e \, (k \cdot p) \, \omega}{m^2} \right) + Gm \left(-\frac{D \, m^2 \, \xi^2 \, (k \cdot p) \, \omega^3}{2 \, s} + \right. \\
& \left. \frac{m^2 \, \xi^2 \, (k \cdot p) \, \omega^3}{s} + \frac{1}{3} D \, m^2 \, \xi^2 \, (k \cdot p) \, \omega^2 - \frac{2}{3} m^2 \, \xi^2 \, (k \cdot p) \, \omega^2 - \frac{D \, qp \, \omega}{2 \, m \, s} + \frac{qp \, \omega}{m \, s} + pp \left(\frac{\omega}{m \, s} - \frac{D \, \omega}{2 \, m \, s} \right) \right) \Big) + \\
& J2 \left(-\frac{D \, m \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega^3}{2 \, s} + \frac{3 \, m \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega^3}{2 \, s} + D \, m \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega^2 - 3 \, m \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega^2 + \right. \\
& \frac{i \, D \, e \, GFDt \, pt \, \omega^2}{m \, s} - \frac{3 \, i \, e \, GFDt \, pt \, \omega^2}{m \, s} - \frac{i \, D \, e \, GFDt \, pt \, \omega}{m} + \frac{3 \, i \, e \, GFDt \, pt \, \omega}{m} + \frac{1}{2} i \, D \, e \, \sigma F \, \omega - \\
& \frac{3}{2} i \, e \, \sigma F \, \omega - \frac{2}{3} D \, m \, s \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega + 2 \, m \, s \, \xi^2 \, \gamma \cdot k \, (k \cdot p) \, \omega - \frac{D \, Gt \, pt \, \omega}{m \, s} + \frac{3 \, Gt \, pt \, \omega}{m \, s} - D - \\
& \frac{1}{2} i \, D \, e \, s \, \sigma F + \frac{3}{2} i \, e \, s \, \sigma F + \Phi^2 \left(-\frac{D \, m \, \omega \, \gamma \cdot k \, \xi^2}{2 \, s \, (k \cdot p)} + \frac{3 \, m \, \omega \, \gamma \cdot k \, \xi^2}{2 \, s \, (k \cdot p)} + Gm \left(\frac{D \, m^2 \, \xi^2 \, \omega}{2 \, s \, (k \cdot p)} - \frac{3 \, m^2 \, \xi^2 \, \omega}{2 \, s \, (k \cdot p)} \right) \right) + \\
& \Phi \left(\frac{D \, e \, \omega \, \gamma \cdot k \, (a \cdot p)}{m \, s \, (k \cdot p)} - \frac{3 \, e \, \omega \, \gamma \cdot k \, (a \cdot p)}{m \, s \, (k \cdot p)} + Gm \left(\frac{3 \, e \, \omega \, (a \cdot p)}{s \, (k \cdot p)} - \frac{D \, e \, \omega \, (a \cdot p)}{s \, (k \cdot p)} \right) \right) + \\
& Gp \left(\frac{D \, \omega \, (k \cdot p)}{m^2 \, s} - \frac{3 \, \omega \, (k \cdot p)}{m^2 \, s} \right) + GFDp \left(-\frac{i \, D \, e \, (k \cdot p) \, \omega^2}{m^2 \, s} + \frac{3 \, i \, e \, (k \cdot p) \, \omega^2}{m^2 \, s} + \frac{i \, D \, e \, (k \cdot p) \, \omega}{m^2} - \frac{3 \, i \, e \, (k \cdot p) \, \omega}{m^2} \right) + \\
& Gm \left(\frac{D \, m^2 \, \xi^2 \, (k \cdot p) \, \omega^3}{2 \, s} - \frac{3 \, m^2 \, \xi^2 \, (k \cdot p) \, \omega^3}{2 \, s} - \frac{1}{3} D \, m^2 \, \xi^2 \, (k \cdot p) \, \omega^2 + \right. \\
& \left. m^2 \, \xi^2 \, (k \cdot p) \, \omega^2 + \frac{D \, qp \, \omega}{2 \, m \, s} - \frac{3 \, qp \, \omega}{2 \, m \, s} + pp \left(\frac{D \, \omega}{2 \, m \, s} - \frac{3 \, \omega}{2 \, m \, s} \right) \right) + 3 - \frac{i \, t \, \gamma \cdot k}{m \, s \, (k \cdot p) \, \omega} \Big)
\end{aligned}$$

Next, we recollect all scalar products into covariant notations

After this step the dependence on $\oplus$ in the preexponent should vanish

$$\text{GFDp pm} = \gamma^\beta \gamma^5 (F^* p)_\beta + \text{GFDt pt}$$

$$\text{Gp pm} = -\text{Gm pp} + (\gamma p) + \text{GFDt pt}$$

$$\text{Gm} = (\gamma k) / m = (\gamma F^2 p) / m [-a^2 (kp)]$$

Note that after this step the preexponential does not depend on $\oplus$

```
In[73]:= SubstitutionStep91 = {DiracGamma[Momentum[k, D], D] (kp) →
  DiracGamma[Momentum[FFp, D], D] e^2 / m^2 / ξ^2,
  GFDp → m / kp (GFDt pt + DiracGamma[LorentzIndex[β, D], D].DiracGamma[5] ×
    Pair[LorentzIndex[β, D], Momentum[FDp, D]]),
  Gp → m / kp (DiracGamma[Momentum[p, D], D] - Gm pp + Gt pt)}
SubstitutionStep92 = {Gm → DiracGamma[Momentum[k, D], D] / m}
SubstitutionStep93 = {DiracGamma[Momentum[k, D], D] →
  DiracGamma[Momentum[FFp, D], D] e^2 / m^2 / ξ^2 / (kp)}
```

$$\text{Out[73]} = \left\{ \gamma \cdot k (k \cdot p) \rightarrow \frac{e^2 \gamma \cdot \text{FFp}}{m^2 \xi^2}, \text{GFDp} \rightarrow \frac{m (\text{FDp}^\beta \gamma^\beta \bar{\gamma}^5 + \text{GFDt pt})}{k \cdot p}, \text{Gp} \rightarrow \frac{m (-\text{Gm pp} + \text{Gt pt} + \gamma \cdot p)}{k \cdot p} \right\}$$

$$\text{Out[74]} = \left\{ \text{Gm} \rightarrow \frac{\gamma \cdot k}{m} \right\}$$

$$\text{Out[75]} = \left\{ \gamma \cdot k \rightarrow \frac{e^2 \gamma \cdot \text{FFp}}{m^2 \xi^2 (k \cdot p)} \right\}$$

```
In[76]:= Coeff9 = Coeff8
Phase9 = Phase8
Matrix9 =
  Collect[
    Expand[
      Simplify[
        Expand[
          Matrix8 /. SubstitutionStep91 /. SubstitutionStep92 /.
            SubstitutionStep93 /. {ω → s t / (s + t)}
        ]
      ]
    ],
    {J0, J1, J2, dJ0dχl, dJ1dχl, dJ2dχl, ⊕, e^2 DiracGamma[Momentum[FFp, D], D],
      DiracGamma[LorentzIndex[β, D], D].DiracGamma[5] ×
        Pair[LorentzIndex[β, D], Momentum[FDp, D]],
      DiracGamma[Momentum[p, D], D], Pair[Momentum[a, D], Momentum[p, D]]}]
```

$$\text{Out[76]} = \frac{1}{4} i e^{-\frac{1}{4} i \pi D} \pi^{\frac{D}{2}-2} e^2 \left(\frac{s t}{\omega} \right)^{-D/2}$$

$$\begin{aligned}
\text{Out}[77] = & \frac{1}{3} m^2 \xi^2 \omega^3 (k \cdot p)^2 - \frac{1}{3} m^2 \xi^2 s \omega^2 (k \cdot p)^2 + \frac{\text{pp} \omega (k \cdot p)}{m} + \frac{\text{qp} \omega (k \cdot p)}{m} - m^2 s + \Phi \left(\frac{\text{qp}}{m} - \frac{\text{pp}}{m} \right) - \text{pt}^2 \omega \\
\text{Out}[78] = & \text{dJ1d}\chi^1 \gamma \cdot \text{FFp} \left(-\frac{i s}{2 m^5 (s+t)^2 \xi (k \cdot p)} - \frac{i t}{2 m^5 (s+t)^2 \xi (k \cdot p)} \right) e^2 + \\
& \text{dJ2d}\chi^1 \gamma \cdot \text{FFp} \left(-\frac{i D s}{2 m^5 (s+t)^2 \xi (k \cdot p)} + \frac{3 i s}{2 m^5 (s+t)^2 \xi (k \cdot p)} - \frac{i D t}{2 m^5 (s+t)^2 \xi (k \cdot p)} + \frac{3 i t}{2 m^5 (s+t)^2 \xi (k \cdot p)} \right) e^2 + \\
& \text{J1} \left(\frac{i e \sigma F s^3}{2 (s+t)^2} + \frac{3 i e t \sigma F s^2}{2 (s+t)^2} - \frac{s^2}{(s+t)^2} + \frac{i e t^2 \sigma F s}{(s+t)^2} - \frac{2 t s}{(s+t)^2} + \left(\frac{t^2}{m (s+t)^2} + \frac{s t}{m (s+t)^2} \right) \gamma \cdot p + \right. \\
& \left. \left(-\frac{i e t s^2}{m (s+t)^2} - \frac{2 i e t^2 s}{m (s+t)^2} \right) \gamma^\beta \cdot \bar{\gamma}^5 \text{FDp}^\beta + e^2 \gamma \cdot \text{FFp} \left(-\frac{2 t s^3}{3 m (s+t)^2} - \frac{2 t^2 s^2}{m (s+t)^2} - \frac{\text{pp} t s}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} + \right. \right. \\
& \left. \left. \frac{\text{qp} t s}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} - \frac{i s}{m^3 (s+t)^2 \xi^2 (k \cdot p)^2} - \frac{\text{pp} t^2}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} + \frac{\text{qp} t^2}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} - \right. \right. \\
& \left. \left. \frac{3 i t}{m^3 (s+t)^2 \xi^2 (k \cdot p)^2} - \frac{3 i t^2}{m^3 (s+t)^2 \xi^2 (k \cdot p)^2 s} - \frac{i t^3}{m^3 (s+t)^2 \xi^2 (k \cdot p)^2 s^2} \right) - \frac{t^2}{(s+t)^2} \right) + \\
& \text{J0} \left(\frac{i D e \sigma F s^3}{2 (s+t)^2} - \frac{2 i e \sigma F s^3}{(s+t)^2} + \frac{i D e t \sigma F s^2}{2 (s+t)^2} - \frac{4 i e t \sigma F s^2}{(s+t)^2} + \frac{D s^2}{(s+t)^2} - \frac{2 i e t^2 \sigma F s}{(s+t)^2} + \frac{2 D t s}{(s+t)^2} + \right. \\
& \left. \left(-\frac{D t^2}{m (s+t)^2} + \frac{2 t^2}{m (s+t)^2} - \frac{D s t}{m (s+t)^2} + \frac{2 s t}{m (s+t)^2} \right) \gamma \cdot p + \left(-\frac{i D e t s^2}{m (s+t)^2} + \frac{6 i e t s^2}{m (s+t)^2} + \frac{4 i e t^2 s}{m (s+t)^2} \right) \gamma^\beta \cdot \bar{\gamma}^5 \text{FDp}^\beta + \right. \\
& \left. e^2 \gamma \cdot \text{FFp} \left(\frac{2 D t s^3}{3 m (s+t)^2} - \frac{4 t s^3}{3 m (s+t)^2} + \frac{4 t^2 s^2}{m (s+t)^2} + \frac{D \text{pp} t s}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} - \right. \right. \\
& \left. \left. \frac{\text{pp} t s}{m^4 (s+t)^2 \xi^2 (k \cdot p)} - \frac{D \text{qp} t s}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} + \frac{\text{qp} t s}{m^4 (s+t)^2 \xi^2 (k \cdot p)} + \frac{D \text{pp} t^2}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} - \right. \right. \\
& \left. \left. \frac{\text{pp} t^2}{m^4 (s+t)^2 \xi^2 (k \cdot p)} - \frac{D \text{qp} t^2}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} + \frac{\text{qp} t^2}{m^4 (s+t)^2 \xi^2 (k \cdot p)} \right) + \frac{D t^2}{(s+t)^2} \right) + \\
& \text{J2} \left(-\frac{i D e \sigma F s^3}{2 (s+t)^2} + \frac{3 i e \sigma F s^3}{2 (s+t)^2} - \frac{i D e t \sigma F s^2}{2 (s+t)^2} + \frac{3 i e t \sigma F s^2}{2 (s+t)^2} - \frac{D s^2}{(s+t)^2} + \frac{3 s^2}{(s+t)^2} - \frac{2 D t s}{(s+t)^2} + \frac{6 t s}{(s+t)^2} + \right. \\
& \left. \left(\frac{D t^2}{m (s+t)^2} - \frac{3 t^2}{m (s+t)^2} + \frac{D s t}{m (s+t)^2} - \frac{3 s t}{m (s+t)^2} \right) \gamma \cdot p + \left(\frac{i D e s^2 t}{m (s+t)^2} - \frac{3 i e s^2 t}{m (s+t)^2} \right) \gamma^\beta \cdot \bar{\gamma}^5 \text{FDp}^\beta + \right. \\
& \left. e^2 \gamma \cdot \text{FFp} \left(-\frac{2 D t s^3}{3 m (s+t)^2} + \frac{2 t s^3}{m (s+t)^2} - \frac{D \text{pp} t s}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} + \frac{3 \text{pp} t s}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} + \right. \right. \\
& \left. \left. \frac{D \text{qp} t s}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} - \frac{3 \text{qp} t s}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} - \frac{i s}{m^3 (s+t)^2 \xi^2 (k \cdot p)^2} - \frac{D \text{pp} t^2}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} + \right. \right. \\
& \left. \left. \frac{3 \text{pp} t^2}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} + \frac{D \text{qp} t^2}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} - \frac{3 \text{qp} t^2}{2 m^4 (s+t)^2 \xi^2 (k \cdot p)} - \frac{3 i t}{m^3 (s+t)^2 \xi^2 (k \cdot p)^2} - \right. \right. \\
& \left. \left. \frac{3 i t^2}{m^3 (s+t)^2 \xi^2 (k \cdot p)^2 s} - \frac{i t^3}{m^3 (s+t)^2 \xi^2 (k \cdot p)^2 s^2} \right) - \frac{D t^2}{(s+t)^2} + \frac{3 t^2}{(s+t)^2} \right)
\end{aligned}$$

```
In[79]:= Coefficient[Matrix9,  $\Phi$ ]
          Coefficient[Matrix9,  $\Phi^2$ ]
```

```
Out[79]= 0
```

```
Out[80]= 0
```

The integration over

$$\int_{-\infty}^{\infty} d\Phi \dots$$

is now trivial,

as the phase is linear in Φ and the preexponent does not depend on Φ

$$\int d\Phi \exp\left[i\Phi \frac{q_+ - p_+}{m}\right] = 2\pi \delta\left(\frac{q_+ - p_+}{m}\right) = 2\pi m \delta(q_+ - p_+)$$

After this step we have collected the full delta - function, so that

$$M(q, p) = \Lambda^{D-4} (2\pi)^D \delta^{(D)}(q - p) M(p)$$

In what follows we will consider $M(p)$

We substitute

$$q = q_+ \rightarrow p = p_+,$$

$$p = \frac{1}{2p_-} (p^2 + p t^2) = \frac{m}{2k p} (p^2 + p t^2)$$

and also introduce

$$\chi = \chi_p = \frac{\xi(kp)}{m^2}$$

After this step all the feasible integrations are done.

We are left with two integrals over the proper times

$$\int_0^\infty ds \int_0^\infty dt \dots$$

and the implicit integration

$$\int_{-\infty}^{\infty} dl^2 \quad \text{in } J_k(t, \chi)$$

In[81]:= **Coeff10 = Coeff9 * 2 π * m / (2 π) ^D / Λ ^ (D - 4)**

Phase10 = Collect[

Expand[

Phase9 /. {qp → pp} /. {pp → (pv2 + pt^2) m / (2 kp)} /. {kp → m^2 χ / ξ }

],

χ , Simplify]

Matrix10 = Collect[Expand[Matrix9 /. {qp → pp} /. {pp → (pv2 + pt^2) m / (2 kp)}] /.

{DiracGamma[Momentum[k, D], D] →

DiracGamma[Momentum[FFp, D], D] e^2 / m^2 / ξ ^2 / (kp)} /. {kp → m^2 χ / ξ },

{J0, J1, J2, dJ0d χ l, dJ1d χ l, dJ2d χ l, e^2 DiracGamma[Momentum[FFp, D], D],

e σ F, DiracGamma[LorentzIndex[β , D], D].DiracGamma[5] ×

Pair[LorentzIndex[β , D], Momentum[FDp, D]], DiracGamma[Momentum[p, D], D],

Pair[Momentum[a, D], Momentum[p, D]], χ], Simplify]

$$\text{Out[81]} = i 2^{-D-1} e^{-\frac{1}{4} i \pi D} \pi^{-\frac{D}{2}-1} e^2 m \Lambda^{4-D} \left(\frac{s t}{\omega} \right)^{-D/2}$$

$$\text{Out[82]} = \frac{1}{3} m^6 \chi^2 \omega^2 (\omega - s) - m^2 s + p^2 \omega$$

$$\begin{aligned} \text{Out[83]} = & J0 \left(-\frac{i e s t \text{FDp}^\beta \gamma^\beta \bar{\gamma}^5 ((D-6)s-4t)}{m(s+t)^2} + \right. \\ & \left. \frac{2 e^2 s^2 t ((D-2)s+6t) \gamma \cdot \text{FFp}}{3 m(s+t)^2} + \frac{i e s \sigma F ((D-4)s-4t)}{2(s+t)} - \frac{(D-2)t \gamma \cdot p}{m(s+t)} + D \right) + \\ & J2 \left(\frac{i (D-3) e s^2 t \text{FDp}^\beta \gamma^\beta \bar{\gamma}^5}{m(s+t)^2} + e^2 \gamma \cdot \text{FFp} \left(-\frac{2(D-3)s^3 t}{3 m(s+t)^2} - \frac{i(s+t)}{m^7 s^2 \chi^2} \right) - \frac{i(D-3) e s^2 \sigma F}{2(s+t)} + \frac{(D-3)t \gamma \cdot p}{m(s+t)} - D + 3 \right) + \\ & J1 \left(-\frac{i e s t \text{FDp}^\beta (s+2t) \gamma^\beta \bar{\gamma}^5}{m(s+t)^2} + e^2 \gamma \cdot \text{FFp} \left(-\frac{2 s^2 t (s+3t)}{3 m(s+t)^2} - \frac{i(s+t)}{m^7 s^2 \chi^2} \right) + \frac{i e s \sigma F (s+2t)}{2(s+t)} + \frac{t \gamma \cdot p}{m s + m t} - 1 \right) - \\ & \frac{i (D-3) \text{dJ2d}\chi\text{l} e^2 \gamma \cdot \text{FFp}}{2 m^7 \chi (s+t)} - \frac{i \text{dJ1d}\chi\text{l} e^2 \gamma \cdot \text{FFp}}{2 m^7 \chi (s+t)} \end{aligned}$$

In[84]:= **Coeff10** /. {D → 4}
Phase10 /. {D → 4}
Matrix10 /. {D → 4}

$$\text{Out[84]} = -\frac{i e^2 m \omega^2}{32 \pi^3 s^2 t^2}$$

$$\text{Out[85]} = \omega \bar{p}^2 + \frac{1}{3} m^6 \chi^2 \omega^2 (\omega - s) - m^2 s$$

$$\begin{aligned} \text{Out[86]} = & -\frac{i \text{dJ1d}\chi_1 e^2 \bar{\gamma} \cdot \overline{\text{FFp}}}{2 m^7 \chi (s+t)} - \frac{i \text{dJ2d}\chi_1 e^2 \bar{\gamma} \cdot \overline{\text{FFp}}}{2 m^7 \chi (s+t)} + \\ & \text{J0} \left(\frac{2 e^2 s^2 t (2s+6t) \bar{\gamma} \cdot \overline{\text{FFp}}}{3 m (s+t)^2} - \frac{i e s t (-2s-4t) \bar{\gamma}^\beta \cdot \bar{\gamma}^5 \overline{\text{FDp}}^\beta}{m (s+t)^2} - \frac{2 t \bar{\gamma} \cdot \bar{p}}{m (s+t)} - \frac{2 i e s \sigma F t}{s+t} + 4 \right) + \\ & \text{J1} \left(e^2 \bar{\gamma} \cdot \overline{\text{FFp}} \left(-\frac{2 s^2 t (s+3t)}{3 m (s+t)^2} - \frac{i (s+t)}{m^7 s^2 \chi^2} \right) - \frac{i e s t (s+2t) \bar{\gamma}^\beta \cdot \bar{\gamma}^5 \overline{\text{FDp}}^\beta}{m (s+t)^2} + \frac{t \bar{\gamma} \cdot \bar{p}}{m s + m t} + \frac{i e s \sigma F (s+2t)}{2 (s+t)} - 1 \right) + \\ & \text{J2} \left(e^2 \bar{\gamma} \cdot \overline{\text{FFp}} \left(-\frac{2 s^3 t}{3 m (s+t)^2} - \frac{i (s+t)}{m^7 s^2 \chi^2} \right) + \frac{i e s^2 t \bar{\gamma}^\beta \cdot \bar{\gamma}^5 \overline{\text{FDp}}^\beta}{m (s+t)^2} + \frac{t \bar{\gamma} \cdot \bar{p}}{m (s+t)} - \frac{i e s^2 \sigma F}{2 (s+t)} - 1 \right) \end{aligned}$$

Let us change the variables :

$$(s, t) \rightarrow (u, \sigma),$$

where

$$s = \frac{1}{m^2} \frac{1+u}{\chi^{2/3} u^{1/3}} \sigma$$

$$t = \frac{1}{m^2} \frac{1+u}{\chi^{2/3} u^{4/3}} \sigma$$

then

$$\omega = \frac{1}{m^2} \frac{1}{\chi^{2/3} u^{1/3}} \sigma,$$

$$\int_0^\infty dt \int_0^\infty ds \dots = \int_0^\infty du \int_0^\infty d\sigma \mid \mathcal{J}^{-1} \mid \dots$$

$$\mid \mathcal{J}^{-1} \mid = \frac{\sigma (u+1)^2}{m^4 u^{8/3} \chi^{4/3}}$$


```

In[87]:= $Assumptions = {χ > 0, u > 0, σ > 0};
suσ = m^(-2) (1 + u) / u^(1/3) / χ^(2/3) σ
tuσ = m^(-2) (1 + u) / u^(4/3) / χ^(2/3) σ
Jac = Simplify[D[suσ, u] × D[tuσ, σ] - D[suσ, σ] × D[tuσ, u]]
Simplify[(1 / suσ + 1 / tuσ) ^ (-1)]

```

$$\text{Out[88]} = \frac{\sigma(u+1)}{m^2 \sqrt[3]{u} \chi^{2/3}}$$

$$\text{Out[89]} = \frac{\sigma(u+1)}{m^2 u^{4/3} \chi^{2/3}}$$

$$\text{Out[90]} = \frac{\sigma(u+1)^2}{m^4 u^{8/3} \chi^{4/3}}$$

$$\text{Out[91]} = \frac{\sigma}{m^2 \sqrt[3]{u} \chi^2}$$

```

In[92]:= Coeff11 = Simplify[Coeff10 * Jac /. {ω → s t / (s + t)} /. {s → suσ, t → tuσ},
  Assumptions → {χ > 0, u > 0, σ > 0}]
Phase11 = Collect[
  Expand[
    Simplify[
      (Phase10) /. {ω → s t / (s + t)} /. {s → suσ, t → tuσ},
      Assumptions → {χ > 0, u > 0, σ > 0}
    ]
  ], {pv2, σ},
  Simplify]
Matrix11 =
Collect[
  Expand[
    Simplify[
      Matrix10 /. {kp → χ / ξ m^2} /. {ω → s t / (s + t)} /. {s → suσ, t → tuσ},
      Assumptions → {u > 0, σ > 0, χ > 0}
    ]
  ],
  {J0, J1, J2, dJ0dχl, dJ1dχl, dJ2dχl, e^2 DiracGamma[Momentum[FFp, D], D],
   e σ F, DiracGamma[LorentzIndex[β, D], D].DiracGamma[5] ×
   Pair[LorentzIndex[β, D], Momentum[FDp, D]],
   DiracGamma[Momentum[p, D], D], pv2, σ, χ}, Simplify]

```

$$\text{Out[92]} = \frac{i 2^{-D-1} e^{-\frac{1}{4} i \pi D} \pi^{-\frac{D}{2}-1} e^2 m \Lambda^{4-D} \left(\frac{\sigma (u+1)^2}{m^2 u^{4/3} \chi^{2/3}} \right)^{2-\frac{D}{2}}}{\sigma (u+1)^2}$$

$$\text{Out[93]} = \frac{p^2 \sigma}{m^2 \sqrt[3]{u} \chi^2} - \frac{\sigma^3}{3} - \frac{\sigma (u+1)}{\sqrt[3]{u} \chi^2}$$

$$\begin{aligned} \text{Out[94]} = & J0 \left(\sigma \text{FDp}^\beta \gamma^\beta \bar{\gamma}^5 \left(\frac{4 i e}{m^3 \sqrt[3]{u} (u+1)^2 \chi^{2/3}} - \frac{i e u^{2/3} (D u + D - 6 u - 10)}{m^3 (u+1)^2 \chi^{2/3}} \right) + \frac{2 e^2 \sigma^2 ((D-2) u + 6) \gamma \cdot \text{FFp}}{3 m^5 u^{2/3} \chi^{4/3}} + \right. \\ & \left. e \sigma \sigma F \left(\frac{i u^{2/3} (D (u+1)^2 - 4 (u^2 + 3 u + 3))}{2 m^2 (u+1)^2 \chi^{2/3}} - \frac{2 i}{m^2 \sqrt[3]{u} (u+1)^2 \chi^{2/3}} \right) + \frac{(2-D) \gamma \cdot p}{m u + m} + D \right) + \\ & J2 \left(\frac{i (D-3) e \sigma u^{2/3} \text{FDp}^\beta \gamma^\beta \bar{\gamma}^5}{m^3 (u+1) \chi^{2/3}} + e^2 \gamma \cdot \text{FFp} \left(-\frac{2 (D-3) \sigma^2 \sqrt[3]{u}}{3 m^5 \chi^{4/3}} - \frac{i}{m^5 \sigma u^{2/3} \chi^{4/3}} \right) - \right. \\ & \left. \frac{i (D-3) e \sigma \sigma F u^{2/3}}{2 m^2 \chi^{2/3}} + \frac{(D-3) \gamma \cdot p}{m (u+1)} - D + 3 \right) + \\ & J1 \left(\sigma \text{FDp}^\beta \gamma^\beta \bar{\gamma}^5 \left(-\frac{i e u^{2/3} (u+3)}{m^3 (u+1)^2 \chi^{2/3}} - \frac{2 i e}{m^3 \sqrt[3]{u} (u+1)^2 \chi^{2/3}} \right) + e^2 \gamma \cdot \text{FFp} \left(-\frac{2 \sigma^2 (u+3)}{3 m^5 u^{2/3} \chi^{4/3}} - \frac{i}{m^5 \sigma u^{2/3} \chi^{4/3}} \right) + \right. \\ & \left. e \sigma \sigma F \left(\frac{i u^{2/3} (u^2 + 4 u + 5)}{2 m^2 (u+1)^2 \chi^{2/3}} + \frac{i}{m^2 \sqrt[3]{u} (u+1)^2 \chi^{2/3}} \right) + \frac{\gamma \cdot p}{m u + m} - 1 \right) - \\ & \frac{i (D-3) dJ2d\chi l e^2 u^{4/3} \gamma \cdot \text{FFp}}{2 m^5 \sigma (u+1)^2 \sqrt[3]{\chi}} - \frac{i dJ1d\chi l e^2 u^{4/3} \gamma \cdot \text{FFp}}{2 m^5 \sigma (u+1)^2 \sqrt[3]{\chi}} \end{aligned}$$

In the next step we rewrite

$$dJ_k d\chi_l = \frac{\partial}{\partial \chi_l} J_k(t, \chi_l),$$

where $\chi_l = \chi_l(u) = \frac{u}{1+u} \chi$ and $t = t(u) = \frac{1}{m^2} \frac{1+u}{\chi^{2/3} u^{4/3}} \sigma$ as defined above

using integration by parts

First, we may use the equality

$$\frac{\partial}{\partial \chi_l} J_k(t, \chi_l) = (\chi_l'(u))^{-1} \left[\frac{d}{du} J_k(t(u), \chi_l(u)) - \frac{\partial}{\partial t} J_k(t, \chi_l) t'(u) \right].$$

Then, we may integrate the first term by parts, i.e.

$$\int_0^\infty du f(u) \frac{d}{du} J_k(t, \chi_l) = - \int_0^\infty du J_k(t, \chi_l) \frac{d}{du} f(t),$$

where we assumed that $J_k(t, 0) = f(\infty) = 0$.

As for second term, we write

$$\begin{aligned} \frac{\partial}{\partial t} J_k(t, \chi_l) &= \frac{\partial}{\partial t} (-i) \int_{-\infty}^\infty dl^2 D_k(l^2, \chi_l) e^{-il^2 t} = \\ &(-i)^2 \int_{-\infty}^\infty dl^2 l^2 D_k(l^2, \chi_l) e^{-il^2 t} = -i m^2 \tilde{J}_k(t, \chi_l). \end{aligned}$$

We denote

$$J_k t = \tilde{J}_k$$

```
In[95]:= χlu = Simplify[χlϕ0 /. {ω → s t / (s + t)} /. {s → suσ, t → tuσ} /. {kp → χ / ξ m^2}] /.
      {kp → χ / ξ m^2}
dχldu = Simplify[D[χlu, u]]
dtdu = Simplify[D[tuσ, u]]
Coefficient[Matrix11, dJ0dχl] * dJ0dχl +
  Coefficient[Matrix11, dJ1dχl] * dJ1dχl + Coefficient[Matrix11, dJ2dχl] * dJ2dχl
Matrix12t = % /. {dJ0dχl → 1 / dχldu (- (-I m^2 J0t) dtdu),
  dJ1dχl → 1 / dχldu (- (-I m^2 J1t) dtdu), dJ2dχl → 1 / dχldu (- (-I m^2 J2t) dtdu)}
Matrix12d =
  %% /. {dJ0dχl → 1 / dχldu dJ0du, dJ1dχl → 1 / dχldu dJ1du, dJ2dχl → 1 / dχldu dJ2du}

Out[95]=  $\frac{u \chi}{u + 1}$ 

Out[96]=  $\frac{\chi}{(u + 1)^2}$ 

Out[97]=  $-\frac{\sigma(u + 4)}{3 m^2 u^{7/3} \chi^{2/3}}$ 
```

$$\text{Out[98]} = -\frac{i(D-3) \text{dJ2d}\chi l e^2 u^{4/3} \gamma \cdot \text{FFp}}{2 m^5 \sigma (u+1)^2 \sqrt[3]{\chi}} - \frac{i \text{dJ1d}\chi l e^2 u^{4/3} \gamma \cdot \text{FFp}}{2 m^5 \sigma (u+1)^2 \sqrt[3]{\chi}}$$

$$\text{Out[99]} = -\frac{(D-3) e^2 \text{J2t}(u+4) \gamma \cdot \text{FFp}}{6 m^5 u \chi^2} - \frac{e^2 \text{J1t}(u+4) \gamma \cdot \text{FFp}}{6 m^5 u \chi^2}$$

$$\text{Out[100]} = -\frac{i(D-3) \text{dJ2du} e^2 u^{4/3} \gamma \cdot \text{FFp}}{2 m^5 \sigma \chi^{4/3}} - \frac{i \text{dJ1du} e^2 u^{4/3} \gamma \cdot \text{FFp}}{2 m^5 \sigma \chi^{4/3}}$$

In[101]:= **f = Coeff11 * Exp[I Phase11]**

**Matrix12dmod = Collect[Expand[Simplify[-D[Matrix12d * f, u] / f] /.
{dJ0du → J0, dJ1du → J1, dJ2du → J2}], {J0, J1, J2}, Simplify]**

$$\text{Out[101]} = \frac{i 2^{-D-1} \pi^{-\frac{D}{2}-1} e^2 m \Lambda^{4-D} \left(\frac{\sigma(u+1)^2}{m^2 u^{4/3} \chi^{2/3}} \right)^{2-\frac{D}{2}} \exp\left(i \left(\frac{p^2 \sigma}{m^2 \sqrt[3]{u \chi^2}} - \frac{\sigma^3}{3} - \frac{\sigma(u+1)}{\sqrt[3]{u \chi^2}} \right) - \frac{i \pi D}{4}\right)}{\sigma(u+1)^2}$$

$$\text{Out[102]} = \frac{e^2 \text{J1} \gamma \cdot \text{FFp} \left(p^2 \sigma(u+1) + m^2 \left(\sigma(2u^2 + u - 1) - i(D-2)(u-2) \sqrt[3]{u \chi^2} \right) \right)}{6 m^7 \sigma(u+1) \chi^2} +$$

$$\frac{(D-3) e^2 \text{J2} \gamma \cdot \text{FFp} \left(p^2 \sigma(u+1) + m^2 \left(\sigma(2u^2 + u - 1) - i(D-2)(u-2) \sqrt[3]{u \chi^2} \right) \right)}{6 m^7 \sigma(u+1) \chi^2}$$

```

In[103]:= Coeff12 = Coeff11
Phase12 = Phase11
Matrix12 = Collect[
  (Matrix11 /. {dJ0dxl -> 0, dJ1dxl -> 0, dJ2dxl -> 0}) + Matrix12t + Matrix12dmod,
  {J0, J1, J2, J0t, J1t, J2t, e^2 DiracGamma[Momentum[FFp, D], D],
   e σF, DiracGamma[LorentzIndex[β, D], D].DiracGamma[5] ×
   Pair[LorentzIndex[β, D], Momentum[FDp, D]],
   DiracGamma[Momentum[p, D], D], pv2, lv2, σ, χ}, Simplify]

```

$$\text{Out[103]} = \frac{i 2^{-D-1} e^{-\frac{1}{4} i \pi D} \pi^{-\frac{D}{2}-1} e^2 m \Lambda^{4-D} \left(\frac{\sigma (u+1)^2}{m^2 u^{4/3} \chi^{2/3}} \right)^{2-\frac{D}{2}}}{\sigma (u+1)^2}$$

$$\text{Out[104]} = \frac{p^2 \sigma}{m^2 \sqrt[3]{u} \chi^2} - \frac{\sigma^3}{3} - \frac{\sigma (u+1)}{\sqrt[3]{u} \chi^2}$$

$$\begin{aligned} \text{Out[105]} = & J2 \left(\frac{i (D-3) e \sigma u^{2/3} \text{FDp}^\beta \gamma^\beta \bar{\gamma}^5}{m^3 (u+1) \chi^{2/3}} + \right. \\ & e^2 \gamma \cdot \text{FFp} \left(\frac{-\frac{i (D^2-5D+6)(u-2) \sqrt[3]{u} \chi^2}{6 m^5 (u+1) \chi^2} - \frac{i}{m^5 u^{2/3} \chi^{4/3}}}{\sigma} + \frac{(D-3) p^2}{6 m^7 \chi^2} - \frac{2 (D-3) \sigma^2 \sqrt[3]{u}}{3 m^5 \chi^{4/3}} + \frac{(D-3) (2u-1)}{6 m^5 \chi^2} \right) - \\ & \left. \frac{i (D-3) e \sigma \sigma F u^{2/3}}{2 m^2 \chi^{2/3}} + \frac{(D-3) \gamma \cdot p}{m (u+1)} - D + 3 \right) + \\ & J0 \left(-\frac{i e \sigma ((D-6)u-4) \text{FDp}^\beta \gamma^\beta \bar{\gamma}^5}{m^3 \sqrt[3]{u} (u+1) \chi^{2/3}} + \frac{2 e^2 \sigma^2 ((D-2)u+6) \gamma \cdot \text{FFp}}{3 m^5 u^{2/3} \chi^{4/3}} + \frac{i e \sigma \sigma F ((D-4)u-4)}{2 m^2 \sqrt[3]{u} \chi^{2/3}} + \frac{(2-D) \gamma \cdot p}{m u + m} + D \right) + \\ & J1 \left(-\frac{i e \sigma (u+2) \text{FDp}^\beta \gamma^\beta \bar{\gamma}^5}{m^3 \sqrt[3]{u} (u+1) \chi^{2/3}} + e^2 \gamma \cdot \text{FFp} \left(\frac{-\frac{i (D-2)(u-2) \sqrt[3]{u} \chi^2}{6 m^5 (u+1) \chi^2} - \frac{i}{m^5 u^{2/3} \chi^{4/3}}}{\sigma} + \frac{p^2}{6 m^7 \chi^2} - \frac{2 \sigma^2 (u+3)}{3 m^5 u^{2/3} \chi^{4/3}} + \frac{2u-1}{6 m^5 \chi^2} \right) + \right. \\ & \left. \frac{i e \sigma \sigma F (u+2)}{2 m^2 \sqrt[3]{u} \chi^{2/3}} + \frac{\gamma \cdot p}{m u + m} - 1 \right) - \frac{(D-3) e^2 J2t (u+4) \gamma \cdot \text{FFp}}{6 m^5 u \chi^2} - \frac{e^2 J1t (u+4) \gamma \cdot \text{FFp}}{6 m^5 u \chi^2} \end{aligned}$$

Let us rewrite the result using the following notation

$$\text{MFFp} = \frac{e^2 \gamma^\mu \text{FFp}_\mu}{m^5 \chi^2} \left(\frac{\chi}{u} \right)^{2/3};$$

$$\text{MFDp} = \frac{e \gamma^\mu \gamma^5 \text{FDp}_\mu}{m^3 \chi} \left(\frac{\chi}{u} \right)^{1/3}, \quad \text{FDp}_\mu = (F^* p)_\mu;$$

$$\text{M}\sigma\text{F} = \frac{e \sigma^{\mu\nu} F_{\mu\nu}}{m^2 \chi} \left(\frac{\chi}{u} \right)^{1/3};$$

```

In[106]:= Coeff13 = Coeff12
Phase13 = Phase12
Matrix13 =
Collect[
Expand[
Matrix12 /.
{DiracGamma[Momentum[FFp, D], D] → MFFp / e^2 * m^5 χ^(4/3) u^(2/3),
DiracGamma[LorentzIndex[β, D], D].DiracGamma[5] ×
Pair[LorentzIndex[β, D], Momentum[FDp, D]] →
MFDp / e * m^3 χ^(2/3) u^(1/3), σF → MσF / e * m^2 χ^(2/3) u^(1/3) }
],
{J0, J1, J2, dJ0dχl, dJ1dχl, dJ2dχl, MFFp, MFDp, MσF, DiracGamma[
Momentum[p, D], D], Pair[Momentum[a, D], Momentum[p, D]], σ}, Simplify]

```

$$\text{Out[106]} = \frac{i 2^{-D-1} e^{-\frac{1}{4} i \pi D} \pi^{-\frac{D}{2}-1} e^2 m \Lambda^{4-D} \left(\frac{\sigma(u+1)^2}{m^2 u^{4/3} \chi^{2/3}} \right)^{2-\frac{D}{2}}}{\sigma(u+1)^2}$$

$$\text{Out[107]} = \frac{p^2 \sigma}{m^2 \sqrt[3]{u \chi^2}} - \frac{\sigma^3}{3} - \frac{\sigma(u+1)}{\sqrt[3]{u \chi^2}}$$

$$\begin{aligned} \text{Out[108]} = & J2 \left(MFFp \left(-\frac{i((D^2 - 5D + 6)u^2 - 2(D^2 - 5D + 3)u + 6)}{6\sigma(u+1)} + \frac{(D-3)\left(\frac{u}{\chi}\right)^{2/3}(m^2(2u-1) + p^2)}{6m^2} - \frac{2}{3}(D-3)\sigma^2 u \right) + \right. \\ & \left. \frac{(D-3)\gamma \cdot p}{m(u+1)} + \frac{i(D-3)MFDp \sigma u}{u+1} - \frac{1}{2} i(D-3)M\sigma F \sigma u - D + 3 \right) + \\ & J0 \left(\frac{(2-D)\gamma \cdot p}{m u + m} - \frac{iMFDp \sigma((D-6)u - 4)}{u+1} + MFFp \sigma^2 \left(\frac{2}{3}(D-2)u + 4 \right) + \frac{1}{2} iM\sigma F \sigma((D-4)u - 4) + D \right) + \\ & J1 \left(MFFp \left(-\frac{i((D-2)u^2 - 2(D-5)u + 6)}{6\sigma(u+1)} + \frac{\left(\frac{u}{\chi}\right)^{2/3}(m^2(2u-1) + p^2)}{6m^2} - \frac{2}{3}\sigma^2(u+3) \right) + \right. \\ & \left. \frac{\gamma \cdot p}{m u + m} - \frac{iMFDp \sigma(u+2)}{u+1} + \frac{1}{2} iM\sigma F \sigma(u+2) - 1 \right) - \frac{MFFp(u+4)((D-3)J2t + J1t)}{6\sqrt[3]{u \chi^2}} \end{aligned}$$

In order to remove the terms in

Matrix13 that are proportional to $(\text{MFFp } \sigma^{-1})$,
we integrate over σ by parts.

Let us consider the integral

$$\int_0^\infty d\sigma g(\sigma) \sigma^{3-D/2} \text{Exp}[-i\sigma^3 - iz\sigma], \quad 4 - D = \epsilon > 0,$$

where $g(\sigma) = J_{1,2}(t(u, \sigma), \chi_1)$, and

where $g(\sigma)$ satisfies the condition :

$$g(\sigma) \sigma^{1-D/2} \rightarrow 0, \quad \sigma \rightarrow 0; \quad g(\infty) = 0.$$

Then, we may write

$$\begin{aligned} \int_0^\infty d\sigma g(\sigma) \sigma^{3-D/2} \text{Exp}[-i\sigma^3 - iz\sigma] &= \\ &= i \int_0^\infty d(\text{Exp}[-i\sigma^3]) g(\sigma) \sigma^{1-D/2} \text{Exp}[-i\sigma^3 - iz\sigma] = \\ &= -i \int_0^\infty d\sigma \sigma^{1-D/2} \text{Exp}[-i\sigma^3 - iz\sigma] \times \left[g'(\sigma) + \left(-\frac{D/2-1}{\sigma} - iz \right) g(\sigma) \right], \end{aligned}$$

therefore

$$\begin{aligned} \int_0^\infty d\sigma \sigma^{1-D/2} \text{Exp}[-i\sigma^3 - iz\sigma] \frac{1}{\sigma} g(\sigma) &= \\ = \int_0^\infty d\sigma \sigma^{1-D/2} \left[\text{Exp}[-i\sigma^3 - iz\sigma] \times \frac{1}{D/2-1} \left[g'(\sigma) - i(\sigma^2 + z) g(\sigma) \right] \right]. \end{aligned}$$

For $g'(\sigma)$ we have

$$\begin{aligned} g'(\sigma) &= \frac{d}{d\sigma} J_{1,2}(t(u, \sigma), \chi_1) = \\ &= \frac{d}{d\sigma} (-i) \int_{-\infty}^\infty dl^2 D_k(l^2, \chi_1) \text{Exp}\left[-i\left(\frac{u}{\chi}\right)^{2/3} \frac{1+u}{u^2} \frac{l^2}{m^2} \sigma\right] = \\ &= (-i) \int_{-\infty}^\infty dl^2 \left(-i\left(\frac{u}{\chi}\right)^{2/3} \frac{1+u}{u^2} \frac{l^2}{m^2}\right) D_k(l^2, \chi_1) \text{Exp}[-it(u, \sigma) l^2] \\ &= -i\left(\frac{u}{\chi}\right)^{2/3} \frac{1+u}{u^2} \tilde{J}_k. \end{aligned}$$

At this step we substitute J_k and $\tilde{J}_k$ explicitly :

$$J_k = -i \int_{-\infty}^\infty dl^2 D_k(l^2, \chi_1) \text{Exp}[-it(u, \sigma) l^2],$$

$$\tilde{J}_k = -i \int_{-\infty}^\infty dl^2 \frac{l^2}{m^2} D_k(l^2, \chi_1) \text{Exp}[-it(u, \sigma) l^2]$$

and introduce notations

$$D_0 = D_0(l^2),$$

$$D_k = D_k(l^2, \chi_1), \quad k = 1, 2.$$

Outer integration : $\int_0^\infty ds \int_0^\infty dt \int_{-\infty}^\infty dl^2 \dots$

The nonrenormalized diagonal part of the mass operator in D dimensions :

```
In[109]:= Coeff14 = Coeff13
Phase14 = Phase13
Matrix14 = Collect[Expand[Simplify[Matrix13 - Coefficient[Matrix13, 1 / σ] / σ +
  (Coefficient[Matrix13, 1 / σ] * I (-σ - z / σ) * σ / (D / 2 - 1) /.
    {z → (u / χ) ^ (2 / 3) (1 - (pv2 - m^2) / m^2 / u + (1 + u) / u^2 lv2 / m^2)}]
  ]] /. {lv2 J1 → m^2 J1t, lv2 J2 → m^2 J2t}, {J0, J1, J2, J1t, J2t, MFFp,
  MFDp, MσF, DiracGamma[Momentum[p, D], D], pv2, u / χ, σ}, Simplify]
```

$$\text{Out[109]} = \frac{i 2^{-D-1} e^{-\frac{1}{4} i \pi D} \pi^{-\frac{D}{2}-1} e^2 m \Lambda^{4-D} \left(\frac{\sigma (u+1)^2}{m^2 u^{4/3} \chi^{2/3}} \right)^{2-\frac{D}{2}}}{\sigma (u+1)^2}$$

$$\text{Out[110]} = \frac{p^2 \sigma}{m^2 \sqrt[3]{u} \chi^2} - \frac{\sigma^3}{3} - \frac{\sigma (u+1)}{\sqrt[3]{u} \chi^2}$$

$$\begin{aligned} \text{Out[111]} = & J2 \left(MFFp \left(\frac{p^2 ((D^2 - 5D + 6)u^2 - (D^2 - 5D + 2)u + 4)}{2(D-2)m^2 \sqrt[3]{u} (u+1) \chi^{2/3}} - \frac{\sigma^2 ((D^2 - 5D + 6)u^2 + 2u + 2)}{(D-2)(u+1)} + \frac{(D^2 - 5D + 2)u - 4}{2(D-2) \sqrt[3]{u} \chi^{2/3}} \right) + \right. \\ & \left. \frac{(D-3)\gamma \cdot p}{m(u+1)} + \frac{i(D-3)MFDp \sigma u}{u+1} - \frac{1}{2} i(D-3)M\sigma F \sigma u - D + 3 \right) - \frac{J2t MFFp ((D^2 - 5D + 6)u^2 + 4u + 4)}{2(D-2)u^{4/3} \chi^{2/3}} + \\ & J0 \left(\frac{(D-2)\gamma \cdot p}{m u + m} - \frac{i MFDp \sigma ((D-6)u - 4)}{u+1} + MFFp \sigma^2 \left(\frac{2}{3} (D-2)u + 4 \right) + \frac{1}{2} i M\sigma F \sigma ((D-4)u - 4) + D \right) + \\ & J1 \left(MFFp \left(\frac{p^2 ((D-2)u^2 - (D-6)u + 4)}{2(D-2)m^2 \sqrt[3]{u} (u+1) \chi^{2/3}} + \frac{\sigma^2 (2(u^2 + u + 1) - D(u^2 + 2u + 2))}{(D-2)(u+1)} + \frac{(D-6)u - 4}{2(D-2) \sqrt[3]{u} \chi^{2/3}} \right) + \right. \\ & \left. \frac{\gamma \cdot p}{m u + m} - \frac{i MFDp \sigma (u+2)}{u+1} + \frac{1}{2} i M\sigma F \sigma (u+2) - 1 \right) - \frac{J1t MFFp ((D-2)u^2 + 4u + 4)}{2(D-2)u^{4/3} \chi^{2/3}} \end{aligned}$$

```
In[112]:= (*Coeff14=Coeff13/(I)
Phase14=Phase13-lv2 tuσ
Expand[Matrix13]/.
{J0→ D0,J1→ D1,J2→ D2,J0t→ lv2/m^2 D0,J1t→ lv2/m^2 D1,J2t→ lv2/m^2 D2};
Matrix14=Collect[Expand[
Simplify[%-Coefficient[%,1/σ]/σ+(Coefficient[%,1/σ]*I(-σ-z/σ)*σ/(D/2-1)/.
{z→(u/χ)^(2/3)(1-(pv2-m^2)/m^2/u+(1+u)/u^2 lv2/m^2)}]]],
{D0,D1,D2,MFFp,MFDp, MσF, DiracGamma[Momentum[p,D],D],pv2,lv2,χ,σ},
Simplify]*)
```

```
In[113]:= "z = " (u / χ) ^ (2 / 3)
Collect[Expand[-Coefficient[Phase14, σ] * (χ / u) ^ (2 / 3)], {pv2, lv2}, Simplify]
```

$$\text{Out[113]} = z = \left(\frac{u}{\chi} \right)^{2/3} \left(-\frac{p^2}{m^2 u} + \frac{1}{u} + 1 \right)$$

The nonrenormalized diagonal part of the mass operator in D = 4 :


```

In[114]:= Coeff14D4 = Coeff14 * σ /. {D → 4}
Phase14D4 = Phase14 /. {D → 4}
Matrix14D4 = Collect[
  Matrix14 / σ /. {D → 4},
  {J0, J1, J2, J1t, J2t, MFFp, MFDp, MσF, DiracGamma[Momentum[p, D], D] /. {D → 4},
  pv2 /. {D → 4}, lv2 /. {D → 4}, χ, σ}, Simplify]

```

$$\text{Out[114]} = -\frac{i e^2 m}{32 \pi^3 (u+1)^2}$$

$$\text{Out[115]} = \frac{\sigma \bar{p}^2}{m^2 \sqrt[3]{u} \chi^2} - \frac{\sigma^3}{3} - \frac{\sigma (u+1)}{\sqrt[3]{u} \chi^2}$$

$$\begin{aligned} \text{Out[116]} = & J0 \left(-\frac{2 \bar{\gamma} \cdot \bar{p}}{\sigma (m u + m)} + \frac{2 i \text{MFDp} (u+2)}{u+1} + \frac{4}{3} \text{MFFp} \sigma (u+3) - 2 i \text{M}\sigma\text{F} + \frac{4}{\sigma} \right) + \\ & J1 \left(\text{MFFp} \left(\frac{(u^2 + u + 2) \bar{p}^2}{2 m^2 \sigma \sqrt[3]{u} (u+1) \chi^{2/3}} - \frac{\sigma (u^2 + 3 u + 3)}{u+1} + \frac{-u-2}{2 \sigma \sqrt[3]{u} \chi^{2/3}} \right) + \right. \\ & \quad \left. \frac{\bar{\gamma} \cdot \bar{p}}{\sigma (m u + m)} - \frac{i \text{MFDp} (u+2)}{u+1} + \frac{1}{2} i \text{M}\sigma\text{F} (u+2) - \frac{1}{\sigma} \right) + \\ & J2 \left(\text{MFFp} \left(\frac{(u^2 + u + 2) \bar{p}^2}{2 m^2 \sigma \sqrt[3]{u} (u+1) \chi^{2/3}} - \frac{\sigma (u^2 + u + 1)}{u+1} + \frac{-u-2}{2 \sigma \sqrt[3]{u} \chi^{2/3}} \right) + \frac{\bar{\gamma} \cdot \bar{p}}{\sigma (m u + m)} + \frac{i \text{MFDp} u}{u+1} - \frac{1}{2} i \text{M}\sigma\text{F} u - \frac{1}{\sigma} \right) - \\ & \frac{J1t \text{MFFp} (u^2 + 2 u + 2)}{2 \sigma u^{4/3} \chi^{2/3}} - \frac{J2t \text{MFFp} (u^2 + 2 u + 2)}{2 \sigma u^{4/3} \chi^{2/3}} \end{aligned}$$

Renormalization

$$M = M_0 + \delta M$$

where we attribute the term incorporating J_0 to M_0 ,
and which coincides with the 1 –

loop order unrenormalized mass operator,
and the rest to δM ,
which is associated with the polarization loop insertions.

M_0 should be renormalized, δM is finite. Therefore,
we take the limit $D \rightarrow 4$ in δM .

We renormalize M_0 as follows

$$M_0(p, F) \rightarrow [M_0(p, 0)]_{\text{ren}} + [M_0(p, F) - M_0(p, 0)]$$

The second term gives a regular field dependent
part. $[M_0(p, 0)]_{\text{ren}}$ is the renormalized field – free mass operator.

In what follows, we write only the field - dependent part of M

Let us introduce the Ritus functions

$$f(z) = i \int_0^\infty d\sigma \exp\left(-i \frac{\sigma^3}{3} - i z \sigma\right),$$

$$f'(z) = \int_0^\infty d\sigma \sigma \exp\left(-i \frac{\sigma^3}{3} - i z \sigma\right),$$

$$f_1(z) = \int_0^\infty \frac{d\sigma}{\sigma} e^{-i z \sigma} \left[\exp\left(-i \frac{\sigma^3}{3}\right) - 1 \right]$$

$$z = \left(\frac{u}{\chi}\right)^{2/3} \left(1 - \frac{1}{u} \frac{p^2 - m^2}{m^2} + \frac{1+u}{u^2} \frac{l^2}{m^2}\right)$$

In effect, in the limit $D \rightarrow$

4 the renormalization of M_0 is reduced to the substitution

$$\frac{1}{\sigma} \exp\left(-i \frac{\sigma^3}{3} - i z \sigma\right) \rightarrow \frac{1}{\sigma} e^{-i z \sigma} \left[\exp\left(-i \frac{\sigma^3}{3}\right) - 1 \right]$$

or

$$\int_0^\infty \frac{d\sigma}{\sigma} \exp\left(-i \frac{\sigma^3}{3} - i z \sigma\right) \rightarrow f_1(z)$$

We also substitute

$$\int_0^\infty d\sigma \sigma \exp\left(-i \frac{\sigma^3}{3} - i z \sigma\right) \rightarrow f'(z)$$

$$\int_0^\infty d\sigma \exp\left(-i \frac{\sigma^3}{3} - i z \sigma\right) \rightarrow -i f(z)$$

and use

$$J_0 = 2 \pi i \Theta(t)$$

Matrix15M0 now contains the phase factor inside the Ritus f - functions.

```
In[117]:= pv2D4 = pv2 /. {D -> 4};
lv2D4 = lv2 /. {D -> 4};
zarg = Collect[Phase14D4 /. {σ^3 -> 0} /. {σ -> -1} /. {1/√[3]{u χ^2} -> (u/χ)^(2/3)/u,
  1/(u^(4/3) χ^(2/3)) -> (u/χ)^(2/3)/u^2}, {(u/χ)^(2/3)}, Simplify]
Clear[f];
Coeff15M0 = Coeff14D4 * 2 π I * 2 /. {e^2 -> α 4 π}
Matrix15M0 =
  Collect[Expand[Coefficient[Matrix14D4, J0] / 2] /. {1/σ -> f1[z], σ -> f'[z],
    MFDp -> -I MFDp f[z], MσF -> -I MσF f[z]}, {f[z], f'[z], f1[z]}]
```

$$\text{Out[119]} = \left(\frac{u}{\chi}\right)^{2/3} \left(-\frac{\vec{p}^2}{m^2 u} + \frac{1}{u} + 1\right)$$

$$\text{Out}[121]= \frac{\alpha m}{2 \pi (u+1)^2}$$

$$\text{Out}[122]= f1(z) \left(2 - \frac{\bar{\gamma} \cdot \bar{p}}{m u + m} \right) + \left(\frac{2 \text{MFFp } u}{3} + 2 \text{MFFp} \right) f'(z) + f(z) \left(\frac{\text{MFDp } u}{u+1} + \frac{2 \text{MFDp}}{u+1} - \text{M}\sigma\text{F} \right)$$

The nontrivial part of δM

$$\text{In}[123]:= \text{Coeff15}\delta M = \text{Coeff14D4} /. \{e^2 \rightarrow \alpha 4 \pi\}$$

$$\text{Phase15} = \text{Phase14D4}$$

$$\text{Matrix15}\delta M = \text{Matrix14D4} - \text{Coefficient}[\text{Matrix14D4}, \text{J0}] \text{J0}$$

$$\text{Out}[123]= -\frac{i \alpha m}{8 \pi^2 (u+1)^2}$$

$$\text{Out}[124]= \frac{\sigma \bar{p}^2}{m^2 \sqrt[3]{u} \chi^2} - \frac{\sigma^3}{3} - \frac{\sigma (u+1)}{\sqrt[3]{u} \chi^2}$$

$$\begin{aligned} \text{Out}[125]= & \text{J1} \left(\text{MFFp} \left(\frac{(u^2 + u + 2) \bar{p}^2}{2 m^2 \sigma \sqrt[3]{u} (u+1) \chi^{2/3}} - \frac{\sigma (u^2 + 3 u + 3)}{u+1} + \frac{-u-2}{2 \sigma \sqrt[3]{u} \chi^{2/3}} \right) + \right. \\ & \left. \frac{\bar{\gamma} \cdot \bar{p}}{\sigma (m u + m)} - \frac{i \text{MFDp} (u+2)}{u+1} + \frac{1}{2} i \text{M}\sigma\text{F} (u+2) - \frac{1}{\sigma} \right) + \\ & \text{J2} \left(\text{MFFp} \left(\frac{(u^2 + u + 2) \bar{p}^2}{2 m^2 \sigma \sqrt[3]{u} (u+1) \chi^{2/3}} - \frac{\sigma (u^2 + u + 1)}{u+1} + \frac{-u-2}{2 \sigma \sqrt[3]{u} \chi^{2/3}} \right) + \frac{\bar{\gamma} \cdot \bar{p}}{\sigma (m u + m)} + \frac{i \text{MFDp } u}{u+1} - \frac{1}{2} i \text{M}\sigma\text{F } u - \frac{1}{\sigma} \right) - \\ & \frac{\text{J1t MFFp} (u^2 + 2 u + 2)}{2 \sigma u^{4/3} \chi^{2/3}} - \frac{\text{J2t MFFp} (u^2 + 2 u + 2)}{2 \sigma u^{4/3} \chi^{2/3}} \end{aligned}$$

Let us rewrite the answer in the following form

$$\begin{aligned} M(p, F) = \sum_{n=0}^2 \left[m S_n(p^2, \chi) + (\gamma p) V_n^{(1)}(p^2, \chi) + \right. \\ \left. \frac{(\gamma F^2 p)}{m^4 \chi^2} V_n^{(2)}(p^2, \chi) + \frac{(\sigma F)}{m \chi} T_n(p^2, \chi) + \frac{(\gamma F^* p) \cdot \gamma^5}{m^2 \chi} A_n(p^2, \chi) \right] \end{aligned}$$

$$\text{MFFpV2} = \text{MFFp} * m \left(\frac{\chi}{u} \right)^{-2/3} == \frac{e^2 \gamma^\mu \text{FFp}_\mu}{m^4 \chi^2};$$

$$\text{MFDpA} = \text{MFDp} * m \left(\frac{\chi}{u} \right)^{-1/3} == \frac{e \gamma^\mu \gamma^5 \text{FDp}_\mu}{m^2 \chi}, \quad \text{FDp}_\mu == (F^* p)_\mu;$$

$$\text{M}\sigma\text{F} = \text{M}\sigma\text{F} * m \left(\frac{\chi}{u} \right)^{-1/3} == \frac{e \sigma^{\mu\nu} F_{\mu\nu}}{m \chi};$$

```

In[126]:= Coeff15M01 = Collect[Matrix15M0 * Coeff15M0 /. {MFFp → MFFpV2 / m (χ / u) ^ (2 / 3),
  MFDp → MFDpA / m (χ / u) ^ (1 / 3), MσF → MσFT / m (χ / u) ^ (1 / 3)},
  {D0, f[z], f'[z], f1[z], DiracGamma[Momentum[p]], MFDpA, MσFT}, Simplify]
Coeff15δM1 =
  Collect[Expand[Matrix15δM * Coeff15δM] /. {MFFp → MFFpV2 / m (χ / u) ^ (2 / 3),
    MFDp → MFDpA / m (χ / u) ^ (1 / 3), MσF → MσFT / m (χ / u) ^ (1 / 3)},
    {MFFpV2, J1, J1t, J2, J2t, DiracGamma[Momentum[p]], MFDpA, MσFT, σ}, Simplify]

```

$$\text{Out[126]} = f_1(z) \left(\frac{\alpha m}{\pi (u+1)^2} - \frac{\alpha \bar{\gamma} \cdot \bar{p}}{2\pi (u+1)^3} \right) + \frac{\alpha \text{MFFpV2} (u+3) \left(\frac{\chi}{u}\right)^{2/3} f'(z)}{3\pi (u+1)^2} + f(z) \left(\frac{\alpha \text{MFDpA} (u+2) \sqrt[3]{\frac{\chi}{u}}}{2\pi (u+1)^3} - \frac{\alpha \text{M}\sigma\text{FT} \sqrt[3]{\frac{\chi}{u}}}{2\pi (u+1)^2} \right)$$

$$\begin{aligned} \text{Out[127]} = & \text{MFFpV2} \left(J_1 \left(\frac{i\alpha (m^2 (u^2 + 3u + 2) - (u^2 + u + 2) \bar{p}^2)}{16\pi^2 m^2 \sigma u (u+1)^3} + \frac{i\alpha \sigma (u^2 + 3u + 3) \left(\frac{\chi}{u}\right)^{2/3}}{8\pi^2 (u+1)^3} \right) + \right. \\ & J_2 \left(\frac{i\alpha (m^2 (u^2 + 3u + 2) - (u^2 + u + 2) \bar{p}^2)}{16\pi^2 m^2 \sigma u (u+1)^3} + \frac{i\alpha \sigma (u^2 + u + 1) \left(\frac{\chi}{u}\right)^{2/3}}{8\pi^2 (u+1)^3} \right) + \\ & \left. \frac{i\alpha J_{1t} (u^2 + 2u + 2)}{16\pi^2 \sigma u^2 (u+1)^2} + \frac{i\alpha J_{2t} (u^2 + 2u + 2)}{16\pi^2 \sigma u^2 (u+1)^2} \right) + \\ & J_1 \left(-\frac{i\alpha \bar{\gamma} \cdot \bar{p}}{8\pi^2 \sigma (u+1)^3} + \frac{i\alpha m}{8\pi^2 \sigma (u+1)^2} - \frac{\alpha \text{MFDpA} (u+2) \sqrt[3]{\frac{\chi}{u}}}{8\pi^2 (u+1)^3} + \frac{\alpha \text{M}\sigma\text{FT} (u+2) \sqrt[3]{\frac{\chi}{u}}}{16\pi^2 (u+1)^2} \right) + \\ & J_2 \left(-\frac{i\alpha \bar{\gamma} \cdot \bar{p}}{8\pi^2 \sigma (u+1)^3} + \frac{i\alpha m}{8\pi^2 \sigma (u+1)^2} + \frac{\alpha \text{MFDpA} \sqrt[3]{u^2 \chi}}{8\pi^2 (u+1)^3} - \frac{\alpha \text{M}\sigma\text{FT} \sqrt[3]{u^2 \chi}}{16\pi^2 (u+1)^2} \right) \end{aligned}$$

```

In[128]:= "S0 = " TraditionalForm[
  Coeff15M01 / m /. {MFFpV2 → 0, DiracGamma[Momentum[p]] → 0, MFDpA → 0, MσFT → 0} ]
"V0(1) = " TraditionalForm[ Coefficient[ Coeff15M01, DiracGamma[Momentum[p]] ] ]
"V0(2) = " TraditionalForm[ (- 8 π^2 / α (1 + u)^2 / (-I)^2 )^(-1) Collect[Expand[
  Coefficient[ Coeff15M01 * (- 8 π^2 / α (1 + u)^2 / (-I)^2 ), MFFpV2]],
  {σ, J1, J1t, J2, J2t, (pv2 /. {D → 4})}], Simplify] ]
"T0 = "
TraditionalForm[(16 π^2 χ / α (χ / u)^(-1/3) (1 + u)^2)^(-1) Collect[Expand[
  Coefficient[ Coeff15M01, MσFT] * (16 π^2 χ / α (χ / u)^(-1/3) (1 + u)^2)],
  {J1, J1t, J2, J2t, σ, (pv2 /. {D → 4})}], Simplify]]
"A0 = " TraditionalForm[ (-8 π^2 χ / α (χ / u)^(-1/3) (1 + u)^3)^(-1) Collect[
  Expand[Coefficient[ Coeff15M01, MFDpA] (-8 π^2 χ / α (χ / u)^(-1/3) (1 + u)^3)],
  {σ, J1, J1t, J2, J2t, (pv2 /. {D → 4})}], Simplify]]

```

$$\text{Out[128]= } S_0 = \frac{\alpha f_l(z)}{\pi (u+1)^2}$$

$$\text{Out[129]= } V_0^{(1)} = \left(-\frac{\alpha f_l(z)}{2\pi (u+1)^3} \right)$$

$$\text{Out[130]= } V_0^{(2)} = \frac{(u+3)\alpha \left(\frac{\chi}{u}\right)^{2/3} f'(z)}{3\pi (u+1)^2}$$

$$\text{Out[131]= } T_0 = \left(-\frac{\alpha \sqrt[3]{\frac{\chi}{u}} f(z)}{2\pi (u+1)^2} \right)$$

$$\text{Out[132]= } A_0 = \frac{(u+2)\alpha \sqrt[3]{\frac{\chi}{u}} f(z)}{2\pi (u+1)^3}$$

```

In[133]:= "S1,2 = " TraditionalForm[
  Coeff15δM1 / m /. {MFFpV2 → 0, DiracGamma[Momentum[p]] → 0, MFDpA → 0, MσFT → 0} ]
"V1,2(1) = " TraditionalForm[ Coefficient[ Coeff15δM1, DiracGamma[Momentum[p]] ] ]
"V1,2(2) = " TraditionalForm[ (- 8 π^2 / α (1 + u)^2 / (-I)^2 )^(-1) Collect[Expand[
  Coefficient[ Coeff15δM1 * (- 8 π^2 / α (1 + u)^2 / (-I)^2 ), MFFpV2]],
  {σ, J1, J1t, J2, J2t, (pv2 /. {D → 4})}], Simplify] ]
"T1,2 = "
TraditionalForm[(16 π^2 / α (χ / u)^(-1/3) (1 + u)^2)^(-1) Collect[Expand[
  Coefficient[ Coeff15δM1, MσFT] * (16 π^2 / α (χ / u)^(-1/3) (1 + u)^2)],
  {J1, J1t, J2, J2t, σ, (pv2 /. {D → 4})}], Simplify]]
"A1,2 = " TraditionalForm[ (-8 π^2 / α (χ / u)^(-1/3) (1 + u)^3)^(-1) Collect[
  Expand[Coefficient[ Coeff15δM1, MFDpA] (-8 π^2 / α (χ / u)^(-1/3) (1 + u)^3)],
  {σ, J1, J1t, J2, J2t, (pv2 /. {D → 4})}], Simplify]]

```

$$\text{Out[133]= } S_{1,2} = \frac{\frac{i J_1 m \alpha}{8 \pi^2 (u+1)^2 \sigma} + \frac{i J_2 m \alpha}{8 \pi^2 (u+1)^2 \sigma}}{m}$$

$$\text{Out[134]= } V_{1,2}^{(1)} = \left(-\frac{i J_1 \alpha}{8 \pi^2 (u+1)^3 \sigma} - \frac{i J_2 \alpha}{8 \pi^2 (u+1)^3 \sigma} \right)$$

$$\text{Out[135]= } V_{1,2}^{(2)} = \frac{i \alpha \left(\sigma \left(\frac{2 J_2 \left(\frac{\chi}{u} \right)^{2/3} (u^2 + u + 1)}{u + 1} + \frac{2 J_1 (u^2 + 3 u + 3) \left(\frac{\chi}{u} \right)^{2/3}}{u + 1} \right) + \frac{\frac{J_1 (u^2 + 2 u + 2)}{u^2} + \frac{J_2 (u^2 + 2 u + 2)}{u^2} + J_1 \left(\frac{u + 2}{u} - \frac{(u^2 + u + 2) \bar{p}^2}{m^2 u (u + 1)} \right) + J_2 \left(\frac{u + 2}{u} - \frac{(u^2 + u + 2) \bar{p}^2}{m^2 u (u + 1)} \right)}{\sigma} \right)}{16 \pi^2 (u + 1)^2}$$

$$\text{Out[136]= } T_{1,2} = \frac{(J_1 (u + 2) - J_2 u) \alpha \sqrt[3]{\frac{\chi}{u}}}{16 \pi^2 (u + 1)^2}$$

$$\text{Out[137]= } A_{1,2} = \left(-\frac{(J_1 (u + 2) - J_2 u) \alpha \sqrt[3]{\frac{\chi}{u}}}{8 \pi^2 (u + 1)^3} \right)$$

The elastic scattering amplitude $\mathcal{M} = \bar{u}_p M(p) u_p$

We assume that $p^2 = 0$ and calculate the matrix element

$$\mathcal{M} = \bar{u}_p M(p) u_p,$$

where u_p is a free Dirac bispinor and $\bar{u}_p = \gamma^0 u_p^\dagger$.

To perform this calculation, we use the following relations

$$\bar{u}_p u_p = 2m,$$

$$\bar{u}_p (\gamma p) u_p = 2m^2,$$

$$\bar{u}_p e^2 (\gamma F^2 p) u_p = 2m^6 \chi^2,$$

$$\bar{u}_p e (\sigma_{\mu\nu} F^{\mu\nu}) u_p = \frac{2}{m} e (\bar{u}_p \gamma^\beta \gamma^5 u_p) (F^* p)_\beta = 4s^\beta e (F^* p)_\beta,$$

where $s^\beta = \frac{1}{2m} \bar{u}_p \gamma^\beta \gamma^5 u_p$ is the electron spin 4 - vector.

Recall

$$MFFp = \frac{e^2 \gamma^\mu F F p_\mu}{m^5 \chi^2} \left(\frac{\chi}{u} \right)^{2/3},$$

$$MFDp = \frac{e \gamma^\mu \gamma^5 F D p_\mu}{m^3 \chi} \left(\frac{\chi}{u} \right)^{1/3}, \quad F D p_\mu = (F^* p)_\mu,$$

$$M\sigma F = \frac{e \sigma^{\mu\nu} F_{\mu\nu}}{m^2 \chi} \left(\frac{\chi}{u} \right)^{1/3},$$

in effect, we should perform the following substitutions

$$\bar{u}_p MFFp u_p = 2m \left(\frac{\chi}{u} \right)^{2/3},$$

$$\bar{u}_p MFDp u_p = \frac{2e}{m^2 \chi} s^\nu F D p_\nu \left(\frac{\chi}{u} \right)^{1/3},$$

$$\bar{u}_p M\sigma F u_p = \frac{4e}{m^2 \chi} s^\nu F D p_\nu \left(\frac{\chi}{u} \right)^{1/3}.$$

```
In[138]:= Simplify[S + DiracGamma[Momentum[p]] / m V1 +
  MFFp V2 (χ / u) ^ (-2 / 3) + MσF T (χ / u) ^ (-1 / 3) + MFDp A (χ / u) ^ (-1 / 3) /.
  {DiracGamma[Momentum[p]] → 2 m^2, MFFp → 2 m χ ^ (2 / 3) / u ^ (2 / 3),
  MσF → e Contract[FVD[s, v] × FDpv[v]] 4 / m^2 / χ * (χ / u) ^ (1 / 3),
  MFDp → e Contract[FVD[s, v] × FDpv[v]] 2 / m^2 / χ * (χ / u) ^ (1 / 3), S → 2 m S}]

Out[138]= 2 e (A + 2 T) (FDp · s)
          m^2 χ
+ 2 m (S + V1 + V2)
```

```

In[139]:= Coeff16M0 = Coeff15M0 * 2 m
zarg0S = zarg /. {(pv2 /. {D → 4}) → m^2}
TermI0 = Expand[Simplify[
  Matrix15M0 /. {DiracGamma[Momentum[p]] → 0, MFFp → 0, MσF → 0, MFDp → 0}]]];
1 / 2 / m Expand[Matrix15M0 - TermI0 + 2 m TermI0] /.
  {DiracGamma[Momentum[p]] → 2 m^2, MFFp → 2 m χ^(2 / 3) / u^(2 / 3),
  MσF → e Contract[FVD[s, v] × FDpv[v]] 4 / m^2 / χ * (χ / u)^(1 / 3),
  MFDp → e Contract[FVD[s, v] × FDpv[v]] 2 / m^2 / χ * (χ / u)^(1 / 3)} /. {pv2 →
  m^2} /. {D → 4};
Matrix16M0 = Collect[
  Expand[
    Simplify[% - Coefficient[%, e Pair[Momentum[s], Momentum[FDp]]]
      e Pair[Momentum[s], Momentum[FDp]]]]
  + Simplify[Coefficient[%, e Pair[Momentum[s], Momentum[FDp]]]
    e Pair[Momentum[s], Momentum[FDp]]]
  ,
  {f[z], f'[z], f1[z]}, Simplify]

```

$$\text{Out[139]} = \frac{\alpha m^2}{\pi (u+1)^2}$$

$$\text{Out[140]} = \left(\frac{u}{\chi} \right)^{2/3}$$

$$\text{Out[143]} = -\frac{e f(z) \left(\frac{u}{\chi} \right)^{2/3} (\overline{\text{FDp}} \cdot \vec{s})}{m^3 (u+1)} + \frac{2}{3} (u+3) \left(\frac{\chi}{u} \right)^{2/3} f'(z) + \frac{(2u+1) f1(z)}{u+1}$$


```

In[144]:= Coeff16δM = Coeff15δM * 2 m
Phase16 = Collect[Phase15 /. {pv2D4 → m^2}, σ, Simplify]
TermIδ = Expand[Simplify[
  Matrix15δM /. {DiracGamma[Momentum[p]] → 0, MFFp → 0, MσF → 0, MFDp → 0}]]];
1 / 2 / m Expand[Matrix15δM - TermIδ + 2 m TermIδ] /.
  {DiracGamma[Momentum[p]] → 2 m^2, MFFp → 2 m χ^(2/3) / u^(2/3),
   MσF → e Contract[FVD[s, v] × FDpv[v]] 4 / m^2 / χ * (χ / u)^(1/3),
   MFDp → e Contract[FVD[s, v] × FDpv[v]] 2 / m^2 / χ * (χ / u)^(1/3)} /.
  {pv2D4 → m^2} /. {D → 4};
Matrix16δM = Collect[
  Expand[
    Simplify[% - Coefficient[%, e Pair[Momentum[s], Momentum[FDp]]]
      e Pair[Momentum[s], Momentum[FDp]]]]
  + Simplify[Coefficient[%, e Pair[Momentum[s], Momentum[FDp]]]
    e Pair[Momentum[s], Momentum[FDp]]]
  ,
  {J1, J2, J1t, J2t, MFFp, MFDp, MσF, DiracGamma[Momentum[p]], χ, σ}, Simplify]

```

$$\text{Out[144]} = -\frac{i \alpha m^2}{4 \pi^2 (u+1)^2}$$

$$\text{Out[145]} = -\frac{\sigma^3}{3} - \sigma \left(\frac{u}{\chi} \right)^{2/3}$$

$$\begin{aligned} \text{Out[148]} = & J1 \left(\frac{i e (u+2) \left(\frac{u}{\chi} \right)^{2/3} (\overline{\text{FDp}} \cdot \vec{s})}{m^3 (u+1)} - \frac{1}{\sigma} + \frac{\sigma (-u^2 - 3u - 3) \chi^{2/3}}{u^{2/3} (u+1)} \right) + \\ & J2 \left(-\frac{i e u \left(\frac{u}{\chi} \right)^{2/3} (\overline{\text{FDp}} \cdot \vec{s})}{m^3 (u+1)} - \frac{1}{\sigma} - \frac{\sigma (u^{4/3} + \frac{1}{u^{2/3}} + \sqrt[3]{u}) \chi^{2/3}}{u+1} \right) - \frac{J1t (u^2 + 2u + 2)}{2 \sigma u^2} - \frac{J2t (u^2 + 2u + 2)}{2 \sigma u^2} \end{aligned}$$

Let us rewrite Matrix16M0

we perform an additional transformation of the D_0 term

Let us use the following integral equality (we will prove it below)

$$\int_0^\infty \frac{du}{(1+u)^2} \left[\frac{2(u-2)u}{3(u+1)} \left(\frac{\chi}{u} \right)^{2/3} f'(z_0) + -\frac{2u}{u+1} f_1(z_0) \right] = 0$$

We add this expression to Matrix16M0

After these transformation we arrive

at the expression corresponding to Eq. (11)

in [A.A.Mironov, S.Meuren, A.M.Fedotov PRD 102, 053 005 (2020)].

```
In[149]:= Matrix16M01 = Collect[Matrix16M0 +  $\left(\frac{2(u-2)u}{3(u+1)} (\chi/u)^{(2/3)} + f1[z] \left(-\frac{2u}{u+1}\right)\right),$ 
  {f[z], f'[z], f1[z]}, Simplify]

Out[149]=  $-\frac{ef(z)\left(\frac{u}{\chi}\right)^{2/3}(\overline{\text{FDp}} \cdot \overline{s})}{m^3(u+1)} + \frac{2(2u^2+2u+3)\left(\frac{\chi}{u}\right)^{2/3}f'(z)}{3(u+1)} + \frac{f1(z)}{u+1}$ 
```

Proof of the integral equality

$$I = \int_0^\infty \frac{du}{(1+u)^2} \left[\frac{2(u-2)u}{3(u+1)} \left(\frac{\chi}{u}\right)^{2/3} f'(z) + \left(\frac{1^2}{m^2} - \frac{2u}{u+1}\right) f_1(z) \right] = 0$$

Let us restore the explicit integral form of the Ritus f - functions and integrate the term with $f'(z)$ by parts

$$\begin{aligned} f'(z) &= \int_0^\infty d\sigma \sigma \exp\left(-i\frac{\sigma^3}{3} - iz\sigma\right) = i \int_0^\infty d\left[\exp\left(-i\frac{\sigma^3}{3}\right) - 1\right] \times \frac{1}{\sigma} e^{-iz\sigma} = \\ &= -i \int_0^\infty d\sigma \times \left[\exp\left(-i\frac{\sigma^3}{3}\right) - 1\right] \times \left[-\frac{1}{\sigma^2} - \frac{iz}{\sigma}\right] e^{-iz\sigma}, \end{aligned}$$

where we used the fact that

$$\left[\exp\left(-i\frac{\sigma^3}{3}\right) - 1\right] \times \frac{1}{\sigma} e^{-iz\sigma} = 0 \text{ for } \sigma = 0 \text{ and } \sigma = \infty.$$

Note that after rewriting $f'(z)$,

the factor $\left[\exp\left(-i\frac{\sigma^3}{3}\right) - 1\right]$ will be common in I ,

$$I = \left[\exp\left(-i\frac{\sigma^3}{3}\right) - 1\right] \int_0^\infty du e^{-iz(u)\sigma} [\dots]$$

so we will only write out the expression in $[\dots]$.

We also substitute

$$z(u) = \left(\frac{u}{\chi}\right)^{2/3} \left(\frac{(u+1)\overline{l}^2}{m^2 u^2} + 1\right).$$

```
In[150]:= zarg1 = zarg0S + lv2D4 tuσ / σ
Int = Collect[1 / (1 + u) ^ 2
  (2 (u - 2) u / 3 / (1 + u) (χ / u) ^ (2 / 3) f' [z] + (lv2D4 / m ^ 2 - 2 u / (1 + u)) f1 [z]) / .
  {f' [z] → -I (-1 / σ ^ 2 - I z / σ), f1 [z] → 1 / σ} /. {z → zarg1},
  {σ, lv2D4}, Simplify]

Out[150]=  $\frac{(u+1)\overline{l}^2}{m^2 u^{4/3} \chi^{2/3}} + \left(\frac{u}{\chi}\right)^{2/3}$ 

Out[151]=  $\frac{\frac{(u+4)\overline{l}^2}{3 m^2 u (u+1)^2} - \frac{2 u}{3 (u+1)^2}}{\sigma} + \frac{2 i \sqrt[3]{u} (u-2) \chi^{2/3}}{3 \sigma^2 (u+1)^3}$ 
```

In the next step we prove that in fact this expression is a total derivative

$$I = \left[\text{Exp} \left(-i \frac{\sigma^3}{3} \right) - 1 \right] \int_0^\infty du [\dots] = \left[\text{Exp} \left(-i \frac{\sigma^3}{3} \right) - 1 \right] \int_0^\infty du \frac{d}{du} \left[P(u) e^{-iz(u)\sigma} \right],$$

and $P(0) = P(\infty) = 0$, which makes the statement evident.

To find $P(u)$,

we expand the derivative $\frac{d}{du} [\dots]$ and equal the coefficient of l^2 to the corresponding coefficient in the expression for I .

We find that

$$P(u) = -\frac{i u^{4/3} \chi^{2/3}}{\sigma^2 (u+1)^2}.$$

In the last two lines we check that the result is correct.

$$I = \int_0^\infty \frac{du}{(1+u)^2} \left[\frac{2(u-2)u}{3(u+1)} \left(\frac{\chi}{u} \right)^{2/3} f'(z) + \left(\frac{l^2}{m^2} - \frac{2u}{u+1} \right) f_1(z) \right] =$$

$$= -\frac{i}{\sigma^2} \left[\text{Exp} \left(-i \frac{\sigma^3}{3} \right) - 1 \right] \int_0^\infty du \frac{d}{du} \left[\frac{u^2}{(u+1)^2} \left(\frac{\chi}{u} \right)^{2/3} e^{-iz\sigma} \right] = 0.$$

In[152]:= IntTest = Collect[

Simplify[D[P[u] Exp[-I zarg1 σ], u] / Exp[-I zarg1 σ]], {λ, ν, P'[u], σ}, Simplify]

Out[152]= $P'(u) - \frac{i \sigma P(u) (2 m^2 u^2 - (u + 4) l^2)}{3 m^2 u^{7/3} \chi^{2/3}}$

```

In[153]:= Psol[u] =
  P[u] /. Solve[Coefficient[Int, lv2D4] == Coefficient[IntTest, lv2D4], P[u]] [[1]]
  Collect[(IntTest /. {P[u] → Psol[u], P'[u] → D[Psol[u], u]}),
    {lv2D4, σ}, Simplify]
  Simplify[Int - % /. { $\sqrt[3]{u \chi^2} \rightarrow \chi^{(2/3)} u^{(1/3)}$ }]

```

$$\text{Out[153]} = -\frac{i u^{4/3} \chi^{2/3}}{\sigma^2 + \sigma^2 u^2 + 2 \sigma^2 u}$$

$$\text{Out[154]} = -\frac{\sigma (2 m^2 u^2 - (u + 4) \tilde{l}^2)}{3 m^2 u (\sigma^2 + \sigma^2 u^2 + 2 \sigma^2 u)} - \frac{4 i \sqrt[3]{u} \chi^{2/3}}{3 (\sigma^2 + \sigma^2 u^2 + 2 \sigma^2 u)} + \frac{i u^{4/3} \chi^{2/3} (2 \sigma^2 + 2 \sigma^2 u)}{(\sigma^2 + \sigma^2 u^2 + 2 \sigma^2 u)^2}$$

$$\text{Out[155]} = 0$$